

The Configured Continuum

A Foundational Whitepaper on Fluid Form as a Necessary Consequence of Arrangement

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Abstract

Standard fluid mechanics treats several of its most basic facts as emergent or simply given: the size a vortex “chooses” for its core, the length scale that turbulence “selects,” and the dissipation that “refuses to vanish” as the viscosity is sent to zero. This whitepaper re-frames those facts as *necessary consequences of arrangement*. We take the continuum to be a coarse-grained, jammed configuration of finite-volume particles—a *volume-particle* (VP) arrangement—and we establish, with reproducible evidence, that the form of fluid motion is *configured*: its governing equations, its selected lengths, and its dissipation are determined by the underlying arrangement rather than postulated. Beneath the pillars we first establish—by inheritance from the companion VP whitepaper and by a local reproduction in this bundle—that the substrate itself *is* a fluid: at the isostatic margin $z = 2d$ the relaxed shear modulus vanishes while the bulk modulus stays finite, so the arrangement shears freely, resists compression, and carries a single elastic (sound) speed $c_s^2 = B/\rho$. The argument then rests on four independently reproducible pillars: (i) *structural*—an arrangement with mass and pairwise interaction forces the continuum balance laws as exact identities, with the Navier–Stokes closure following conditionally; (ii) *geometric*—a rotational core capacity identity fixes an effective core length exactly once the arrangement is specified, reproducing a published cross-regime audit to machine precision; (iii) *dynamical*—a binding versus penalty competition selects a length $L_\star \propto (\sigma/\varepsilon)^{1/2}$ with an exact one-half control exponent, independent of the growth offset; and (iv) *dissipative*—rearrangement events carry a nonnegative dissipation measure whose steady-state budget saturates at the injection rate as viscosity vanishes (an Onsager-type anomaly). Every claim is graded **[LOCK]** (locked primitive), **[DERIVE]** (proved/derived), or **[GATE]** (falsifiable, conditional), and every computational claim ships with a runnable box that was executed to the stated tolerance. We respect the empirical phenomena exactly as standard fluid mechanics records them; what changes is their status: they stop being mysteries and become consequences. The framework has already passed its first external test—the same two substrate rules that make the medium flow force the laminar–turbulent transition into the directed-percolation universality class, the class experiment confirmed (2016–2024)—and a second, independent one: the dissipation events of developed turbulence are distributed as marginal-stability avalanches, their thresholded size law matching the pre-registered exponent ($\tau \approx 1.4$ – 1.5) measured directly here; and on release into decay the dissipation coefficient follows the Vassilicos non-equilibrium law $C_\varepsilon \propto \text{Re}_\lambda^{-1}$ before relaxing onto the constant- C_ε plateau — the zeroth law as an RG fixed point. The transition-critical roughness exponent tracks the DP spatial dimension (quasi-2D $2\beta/\nu_\perp = 1.48$ against 1.59), and at the maximum-drag-reduction

bounding line the dissipation-event exponent is polymer-universal. Every claim ships with the scaffolding to attack it.

Contents

How to read this document	4
1 Executive summary	6
2 The reframing: from mystery to necessity	8
2.1 Four facts usually left as “emergent”	8
2.2 The interpretive stance: respect the phenomena, change the explanation	9
2.3 The arrangement postulate	9
3 The marginal substrate: why a jammed arrangement flows	10
3.1 Counting constraints: where $z = 2d$ comes from	10
3.2 The fluidity theorem: zero shear reserve at the margin	10
3.3 Rotation is the unjamming switch	12
3.4 The acoustic face: one speed survives	12
3.5 The constitutive ladder: where each closure lives	12
3.6 Why a driven medium sits at the margin (G-SOC)	13
3.7 The transition test: a 140-year problem lands where the substrate says	13
3.8 What is inherited, what is local, what is not imported	14
4 The axiomatic core: rotation forces a three-body structure	15
4.1 State, invariants, and the two-body kernel	15
4.2 The No-Go theorem: two bodies are not enough	16
4.3 The three-body term and its triangle	17
4.4 Event dissipation: how the arrangement sheds energy	17
4.5 What is proved, and what this whitepaper verifies	18
5 Dimensionless diagnosis: the Π-invariants and the event RG	18
5.1 The Π -invariants	18
5.2 The event renormalization group	19
6 Pillar I — Structural: arrangement forces the equations	19
6.1 From arrangement to fields	20
6.2 Mass and momentum balance are exact	20
6.3 The Newtonian closure is conditional, not automatic	21
6.4 What the solver confirms	21
6.5 The compressible corollary: the arrangement has a sound speed	22
7 Pillar II — Geometric: arrangement fixes length	22
7.1 The rotational core capacity identity (RCCI)	22
7.2 The cross-regime audit and what it reveals	23
8 Pillar III — Dynamical: arrangement selects length	25
8.1 Selection as an extremum, not a cascade	25
8.2 Selection is an attractor, not only an extremum	25
8.3 From Navier–Stokes to the selection normal form	26
8.4 The selection is real in the nonlinear pattern	26

9	Pillar IV — Dissipative: arrangement sets dissipation	27
9.1	Events as a dissipation measure	27
9.2	What the events are: local unjamming of the marginal substrate	27
9.3	The anomaly as channel migration	28
9.4	The metriplectic decomposition, derived	28
9.5	What reproduces, and what is deliberately left open	29
10	Reach and universality: the same mechanism in foreign domains	32
10.1	Dissipation transplanted: Burgers shocks are events	32
10.2	Geometry transplanted: the capacity identity is dimension-general	33
10.3	Selection transplanted: chemistry selects the same length	33
10.4	What the reach test establishes	34
11	Cross-scale extensibility: one arrangement law from genomes to vortices	34
11.1	The same capacity identity in genomes and in vortices	35
11.2	Droplet formation is length selection	35
11.3	Why a planar vortex breathes but a spherical core closes	35
11.4	The co-rotation mechanism: how small rotations build a large one	36
11.5	The extensibility map	39
12	The thesis at work: natural form as forced arrangement	39
12.1	Jupiter’s Great Red Spot: a selected size that persists	40
12.2	Saturn’s hexagon: mode-locking to a discrete symmetry	40
12.3	The typhoon: forced co-rotation made planetary	40
12.4	Convection: plumes as the events that carry the flux	40
12.5	Drag reduction: the marginal bounding line	41
12.6	The granular avalanche: flow as sustained unjamming	41
13	Synthesis and the reproducibility spine	42
13.1	Four faces of one statement	42
13.2	Consolidated reproducibility ledger	42
13.3	The no-tuning audit	42
13.4	The four layers close on each other	43
14	Claims, non-claims, and falsification	43
14.1	What we claim	44
14.2	What we explicitly do <i>not</i> claim	44
14.3	Falsification criteria (per pillar)	45
14.4	Open gates and the research program	45
A	Reproducibility bundle and run instructions	45
B	Algorithm boxes (complete forms)	46
C	The axiomatic-core module (axioms.py)	50
D	The Navier–Stokes solver (ns2d.py)	50
E	The metriplectic vortex-gas model (metriplectic_vortex.py)	51
F	The length-selection solver (length_selection.py)	51
G	The RCCI audit verifier (verify_rotcore.py)	51

H	The universality / reach module (<code>universality.py</code>)	51
I	The cross-scale extensibility module (<code>extensibility.py</code>)	51
J	The rigid-shell / quantum-length module (<code>rigid_shell.py</code>)	51
K	The co-rotation mechanism module (<code>corotation.py</code>)	51
L	The lattice-with-inflow module (<code>lattice_inflow.py</code>)	52
M	The event-flux module (<code>event_flux.py</code>)	52
N	The multidimensional-flux module (<code>multid_flux.py</code>)	52
O	The 3D Navier–Stokes solver (<code>ns3d.py</code>)	52
P	The event-RG module (<code>event_rg.py</code>)	53
Q	The marginal-substrate module (<code>marginal_fluidity.py</code>)	53
R	The unjamming/attractor module (<code>unjam_inflow.py</code>)	53
S	The transition-class module (<code>transition_dp.py</code>)	53
T	The dissipation-avalanche module (<code>dissipation_avalanche.py</code>)	53
U	The non-equilibrium-dissipation module (<code>nonequilibrium_dissipation.py</code>)	54
V	The quasi-2D transition module (<code>transition_dp_2d.py</code>)	54
W	The MDR-universality module (<code>mdr_universality.py</code>)	54
X	Pinned Navier–Stokes reference table	54

How to read this document

Grading legend. Every load-bearing statement carries one of three labels, used in the same spirit as the companion VP physics whitepaper (*VP Theory*, concept DOI [10.5281/zenodo.17932566](https://doi.org/10.5281/zenodo.17932566); cited canonically in Section 3):

- **[LOCK]** — a *locked primitive or definition*. It is adopted as a modeling choice or a bookkeeping convention, not proved. Locks are the inputs; they are stated explicitly so the reader knows exactly what is assumed.
- **[DERIVE]** — a *derived or proved* statement. Given the locks, it follows by mathematics (an identity, a theorem, or a computation reproduced to a stated tolerance). Derivations carry no additional fitting.
- **[GATE]** — a *conditional, falsifiable* statement. It holds only if a declared regime/test is passed, and it can fail. Gates are where the framework exposes itself to refutation; each gate names its own failure condition.

Inherited results and their provenance. A small set of substrate results (Section 3) is *inherited* from the companion whitepaper rather than first established here. The rule is uniform

and conservative: an inherited statement keeps exactly the grade its evidence earns. The companion’s **Forced** maps to **[DERIVE]** when the derivation is self-contained algebra or geometry (re-derived from scratch in the text); its **Verified** maps to **[DERIVE]** whose reproducibility box names the companion bundle and concept DOI *and*, wherever feasible, a local small- N reproduction shipped in *this* bundle; its **Hypothesis** maps to **[GATE]** with the falsification route restated. Anything resting on the companion’s *dimensional anchor* (absolute scales, physical constants) is deliberately *not* imported: Section 3 closes dimensionless. Citations to the companion use the concept DOI only.

Reproducibility boxes. Computational claims appear in teal boxes that name the script, the exact knobs (seed, resolution, step, window), and the measured tolerance. The scripts form a single self-contained bundle that runs on `numpy/scipy` alone, with fixed seeds (deterministic): `ns2d.py`, `metriplectic_vortex.py`, `length_selection.py`, `verify_rotcore.py`, `validate_all.py`, `universality.py`, `extensibility.py`, `rigid_shell.py`, `corotation.py`, `lattice_inflow.py`, `event_flux.py`, `multid_flux.py`, `ns3d.py`, `event_rg.py`, `marginal_fluidity.py`, `unjam_inflow.py`, `transition_dp.py`, `dissipation_avalanche.py`, `nonequilibrium_dissipation.py`, `transition_dp_2d.py`, and `mdr_universality.py`. Each box states which script reproduces it.

Algorithm boxes. Because this document argues a different paradigm, it does not ask the reader to trust the bundle on faith. Every key computational result also carries an *orange algorithm box* giving the input, the numbered steps, the output, and the PASS condition in plain pseudocode. The intent is concrete: a reader—or an AI—should be able to re-implement and run the result from the box alone, *without opening the reproducibility files*, and arrive at the stated tolerance. The boxes in the body are summaries; their complete forms are collected in Appendix B.

The reproducibility map. The table below is the document’s spine: it links each load-bearing claim to the script that reproduces it and the tolerance achieved. It appears here in summary and in full as the master ledger of Section 13. No row depends on a fitted parameter.

Claim	§	Script	Tolerance / result
Axiomatic core (No-Go, U3)	4	<code>axioms.py</code>	central; identity $\leq 10^{-15}$
Pillar I (balance laws)	6	<code>ns2d.py</code> , <code>validate_all.py</code>	invariants $\leq 10^{-5}$
Pillar II (RCCI identity)	7	<code>verify_rotcore.py</code>	machine precision
Pillar III (1/2 law)	8	<code>length_selection.py</code>	$\hat{\beta} \approx 0.501$
Pillar IV (budget, saturation)	9	<code>metriplectic_vortex.py</code>	closes; $\rightarrow 0.998 I$
Pillar IV (event flux)	9	<code>event_flux.py</code>	ν -indep, 100% at shock
Pillar IV (2D NS flux)	9	<code>multid_flux.py</code>	events $\approx 60\%$, res.-stable
Pillar IV (3D NS flux)	9	<code>ns3d.py</code>	strain events $\approx 60\%$, conv.
Pillar IV (event size law)	9	<code>dissipation_avalanche.py</code>	$\tau \rightarrow 1.46 \in [1.4, 1.5]$
Pillar IV (C_ϵ non-equilibrium)	9	<code>nonequilibrium_dissipation.py</code>	$C_\epsilon \propto \text{Re}_\lambda^{-1.0}$; plateau
Transition (quasi-2D roughness)	3.7	<code>transition_dp_2d.py</code>	$2\beta/\nu_\perp = 1.48$ (DP 1.59)
Drag reduction (MDR universality)	12.5	<code>mdr_universality.py</code>	event exponent polymer-invariant
Universality U1–U3	10	<code>universality.py</code>	d -general; 6–8%
Mechanism M (co-rotation)	11.4	<code>corotation.py</code>	frustration, merger, Ekman
G-Q (lattice nozzle)	11	<code>lattice_inflow.py</code>	2D sustains / 3D closes
G-S (rigid-shell birth)	11	<code>rigid_shell.py</code>	spread 3.4% $\rightarrow$ 1.8%
Extensibility E1–E4	11	<code>extensibility.py</code>	identity exact; selections
Event RG (Π_L plateau)	5	<code>event_rg.py</code>	exact at fixed point
Substrate fluidity ($G_{\text{rel}} \rightarrow 0$, B finite, $c^2 = B/\rho$)	3	<code>marginal_fluidity.py</code>	z_0 CI $\ni 2d$; c^2 0.4%
Rotation unjams; forced-radius attractor	3	<code>unjam_inflow.py</code>	corr 0.96–0.99; 10^{-15}
Transition class = DP (five exponents)	3	<code>transition_dp.py</code>	$\delta = 0.158$ vs 0.1595

The one principle that organizes everything. Read every chapter as a different projection of a single sentence:

Organizing thesis

Fluid form follows arrangement. The equations, the geometry, the selected length scales, and the dissipation of a flow are determined by the configuration of the underlying medium. What looks emergent or mysterious in a field description becomes necessary once the arrangement is made explicit.

1 Executive summary

One sentence, verified four ways and now grounded once beneath them: *fluid form follows arrangement*. The substrate result $G_{\text{relaxed}} \rightarrow 0$ with B finite at $z = 2d$ makes the medium a fluid; Pillars I–IV then force its equations, fix and select its lengths ($L_\star \propto (\sigma/\varepsilon)^{1/2}$), and set its dissipation.

This whitepaper does one thing: it replaces the unexamined premise of fluid mechanics—*that there is a fluid*—with a measured structure, and shows that the equations, the geometry, the selected lengths, and the dissipation follow from that structure rather than standing beside it. Four lines of the author’s work turn out to be four faces of this one claim. Each is summarized here and developed in its own chapter.

The reframing (Section 2). Four textbook facts—that the medium *flows* at all, core size, length selection, anomalous dissipation—are usually treated as outputs of a dynamics whose origin is left open. We invert the order of explanation: we posit an *arrangement* (a jammed configuration of finite-volume particles), and show that the four facts are forced by it. The move is deliberately conservative about phenomena and deliberately strong about interpretation.

The marginal substrate (Section 3). Before any pillar there is a zeroth question the field description cannot even pose: *why is the medium a fluid at all?* The answer is inherited from the companion whitepaper and reproduced locally in this bundle. A jammed arrangement at the isostatic margin $z = 2d$ has zero shear reserve—the relaxed shear modulus vanishes, $G_{\text{relaxed}} \propto (z - 2d) \rightarrow 0$, while the bulk modulus stays finite—so it shears freely, resists compression, and carries one elastic (sound) speed $c_s^2 = B/\rho$. Fluidity is not an assumption of A0; it is a measured property of the marginal arrangement (bootstrap z_0 confidence interval containing $2d$; `marginal_fluidity.py`). The same marginality makes *rotation* the unjamming switch: a rigid rotation costs the packing nothing, while any differential rotation is a shear that exceeds the vanishing reserve and breaks contacts (`unjam_inflow.py`). The same two rules, taken alone, fix the universality class of the laminar–turbulent transition—directed percolation, the class experiments confirmed in 2016–2024 (§3.7).

The axiomatic core (Section 4). Underneath the four pillars is a single structural fact about rotating media: a two-body interaction is provably insufficient (a No-Go theorem—a co-rotating pair cannot self-propel), so the minimal interaction is an irreducible *three-body* term whose minimal unit is a triangle. Energy leaves the conservative core only through rearrangement *events*, by the law $\varepsilon_{\text{bind}} = \Theta\sigma$. This core is what the lattice C_3 co-rotation and the entire dissipation story rest on, and its key statements reproduce to machine precision.

Dimensionless diagnosis (Section 5). A selected arrangement is recognized by three dimensionless invariants (Π_T, Π_L, Π_{ST}) lying on a line, and is *explained* by an event renormalization group whose fixed point—where Π_L stops flowing, $(1 + \alpha_\star) = \psi + \chi$ —is the configuration the medium settles into. This is the operational, falsifiable form of the thesis, and it is the same statement the one-half law of Pillar III makes in real space.

Pillar I — Structural: arrangement forces the equations (Section 6). From a VP configuration $\{(\mathbf{x}_i, \mathbf{v}_i, m_i)\}$ and a coarse-graining kernel, the continuity and momentum balances are *exact identities*—the Irving–Kirkwood construction—so the *form* of fluid dynamics is not an assumption but a consequence of having mass and pairwise interaction ([DERIVE]). The Newtonian (Navier–Stokes) closure is then a *conditional* implication: it follows from local isotropic linear response in a declared regime ([GATE]). A pseudo-spectral solver confirms the two exact corollaries that matter—conservation of the inviscid invariants and the energy-budget identity $\dot{E} = -\varepsilon_\nu$ —to 3×10^{-8} and 4×10^{-6} . Its compressible specialization carries the substrate sound speed $c_s^2 = B/\rho$, measured at the margin in Section 3.

Pillar II — Geometric: arrangement fixes length (Section 7). For any rotationally dominated core, the rotational core capacity identity $N_\star r_{\text{eff}}^2 = \phi s L_{\text{core}}^2$ is an exact accounting equality ([LOCK]/[DERIVE]): once the arrangement $(L_{\text{core}}, \phi, s, N_\star)$ is fixed, the effective length is determined. A cross-regime audit (tropical cyclone, ocean eddy, two-dimensional turbulence, multi-vortex surrogate) shows precisely when a single length is an adequate summary and when the arrangement is irreducibly multiscale; the published tables reproduce from the sample-level data to 3×10^{-12} , and the identity itself to 10^{-15} .

Pillar III — Dynamical: arrangement selects length (Section 8). When a binding tendency (k^2) competes with a penalty (k^4), the linear dispersion $\lambda(k) = \mu + \varepsilon k^2 - \sigma k^4$ has a stationary extremum that *selects* $k_\star = \sqrt{\varepsilon/2\sigma}$, hence a selected length $L_\star \propto (\sigma/\varepsilon)^{1/2}$ *independent of the growth offset* μ , with an exact one-half control exponent ([DERIVE]). A nonlinear pattern locks onto $k_\star$ and reproduces the one-half slope; the companion spectrally-rotating DNS recovers $\hat{\beta} \approx 0.501$ with a confidence interval containing $1/2$.

Pillar IV — Dissipative: arrangement sets dissipation (Section 9). Rearrangement events (jamming/unjamming/merger) release binding energy as a nonnegative space–time measure. In a forced steady window, energy conservation forces the budget $\langle \varepsilon_{\text{inj}} \rangle = \langle \varepsilon_\nu \rangle + \langle \varepsilon_{\text{events}} \rangle$ ([DERIVE], exact bookkeeping); as $\nu \rightarrow 0$ the dissipation migrates from the smooth channel into the event channel and *saturates* at the injection rate—an Onsager-type anomaly in which the dissipation is set by the large-scale arrangement, not by the microscopic viscosity. The mean-field budget closes exactly and the saturation is reproduced; the stronger statement that the event channel physically carries the inertial flux to the dissipation scale is *exact in 1D Burgers, resolution-stable in 2D, and resolution-converged in developed 3D turbulence* (the forward energy flux concentrates on strain events, $\text{corr} \approx 0.80$, across $N = 64, 80, 96, 128$), leaving only the infinite-Reynolds asymptotic limit as an explicit [GATE]. At substrate level the events now have a face: they are *local unjamming avalanches* of the marginal arrangement (Section 3).

Reach — the same mechanism in foreign domains (Section 10). With the four pillars in place, we test how far the thesis reaches, by transplanting each mechanism into a domain it was not built for. Event-dissipation reappears in 1D Burgers, where shocks are the events and the dissipation saturates as $\nu \rightarrow 0$; the capacity identity holds in any dimension ($d = 2, 3, 4, 5$ to 10^{-16}); and length selection reappears in a chemical reaction–diffusion system, whose Turing pattern locks onto the dispersion-predicted length. Each mechanism is, in fact, an instance of a

structure physics already treats as universal—the pattern-selection normal form, box-counting geometry, and the anomalous-dissipation defect measure—and A0 is what unifies them.

Extensibility across scale (Section 11). The reach is not only across domains but across scale. An independent line of work on genomes arrives at the *same* capacity identity ($Nv_{\text{eff}} = \phi V_{\text{core}}$, i.e. the RCCI in $d = 3$), the same two knobs (interfacial tension and fill fraction), and the same minimum-residual-stress selection; droplet formation is the Rayleigh–Plateau realization of Pillar III; and a planar vortex breathes while a spherical core closes, by continuity alone. This section also develops *Mechanism M*, the co-rotation chain that the axiomatic triangle predicts: small rotations are frustrated into forced co-rotation, merge into one large rotation, and pump a through-flow—demonstrated, and tested on the 81+1 lattice. Bridges to the quantum core (the 82-lattice; rigid-shell birth setting a length) are stated as graded conjectures with explicit falsification routes.

The thesis at work (Section 12). Five long-studied phenomena are then read as selected arrangements rather than solved from first principles: the Great Red Spot’s size and longevity as one fixed-point fact, Saturn’s hexagon as a locked integer symmetry, the typhoon as Mechanism M at planetary scale, convection as plumes carrying the event flux, and ocean and zonal jets as a selected staircase spacing. The empirical phenomena are untouched; the claim is only that one sentence organizes them.

Synthesis and honesty (Sections 13–14). The four pillars cohere into one statement and one reproducibility spine (Section 13, with a consolidated ledger). We then state plainly what the framework *claims* and what it *does not* (Section 14): we do not overturn the Navier–Stokes equations (we derive them), we do not assert a proof of global regularity (we reframe the question), and we do not claim the VP arrangement is the unique microscopic substrate (it is a sufficient, falsifiable model). Each pillar carries an explicit falsification criterion.

2 The reframing: from mystery to necessity

Four textbook facts—that the medium flows at all, core size, length selection, anomalous dissipation—are usually left as “emergent.” Each is reframed as a necessary consequence of an explicit arrangement; the phenomena are untouched, the explanation changes. The zeroth fact is settled by a measurement: $G_{\text{relaxed}} \rightarrow 0$ at $z = 2d$ while B stays finite.

2.1 Four facts usually left as “emergent”

A great deal of fluid mechanics can be done without ever asking why certain *forms* appear. Four are emblematic.

1. **Fluidity itself.** Before any pattern, the medium *flows*: it yields to the smallest shear yet resists compression and transmits sound. In a field description this is the *definition* of a fluid, hence not explainable within it; in the arrangement description it is a property the configuration must earn—and Section 3 shows the marginal jammed arrangement earns it exactly ($G_{\text{relaxed}} \rightarrow 0$, B finite).
2. **Core size.** A tropical cyclone eye, a mesoscale ocean eddy, and a two-dimensional vortex each present a fairly definite radius. Operationally this radius is measured and used, but in a pure field description there is no *a priori* reason the flow should admit a single characteristic core length at all.

3. **Length selection.** Under suitable forcing, a flow develops structure at a preferred scale. The preferred scale is reproducible, yet in the nonlinear field equations it is not obviously written anywhere; it appears to be “chosen” by the dynamics.
4. **Anomalous dissipation.** As viscosity is reduced, the dissipation rate of a turbulent flow does not follow it to zero but tends to a finite limit. The standard field statement of this (the Onsager picture) is correct and empirically supported, but it is usually presented as a property of the inviscid *limit* rather than as a consequence of anything concrete in the medium.

None of these is a paradox. Each is, in the conventional account, an *emergent* feature: real, measured, and used, but not derived from a stated structure of the medium. The thesis of this whitepaper is that all four become *necessary*—not plausible, necessary—the moment one takes seriously that the medium has an *arrangement*.

2.2 The interpretive stance: respect the phenomena, change the explanation

We are explicit about the epistemic move, because it is the source of the document’s trustworthiness.

Principle 1 (Phenomena are invariant; interpretation is the variable). Every empirical phenomenon used in this whitepaper—measured radii, dissipation curves, selected wavenumbers, the cross-regime audit tables—is taken exactly as standard fluid mechanics records it. We add no data and we contradict no measurement. What we change is the *order of explanation*: instead of treating form as an unexplained output of a dynamics, we treat it as a determined output of an *arrangement*, and we test that determination.

This is why the grading matters. A reader who rejects the VP interpretation entirely can still use every [LOCK] and every [DERIVE] in this document, because the locked primitives are conventions and the derivations are mathematics or reproduced computations. The interpretation lives in how the pillars are *assembled*—in the claim that “configured” is the right word for what the mathematics shows—and in the [GATE] statements, which are offered precisely so they can be attacked.

2.3 The arrangement postulate

The VP arrangement (locked modeling primitive)

[LOCK] (A0). The continuum is the coarse-graining of an *arrangement*: a configuration $\{(\mathbf{x}_i, \mathbf{v}_i, m_i)\}_{i=1}^N$ of finite-volume particles with (i) a well-defined mass m_i , (ii) finite-volume localization (the particles occupy space and can jam), and (iii) a well-defined record of *rearrangement events* (a time, the participants, and a nonnegative released binding energy). A smoothing kernel W_ℓ of width ℓ turns the arrangement into fields $\rho, \mathbf{u}, e$.

We do not claim A0 is the unique microscopic truth of fluids; we claim it is *sufficient*—that from A0 alone, with no further fitting, the four pillars follow at the gradings stated. The remainder of the document is the demonstration. The logical skeleton is uniform: each pillar declares its locks, derives its consequence, and exposes a gate, and each is checked by a script that the reader can run.

A note on what “arrangement” buys that a pure field does not. A field $\mathbf{u}(\mathbf{x}, t)$ is infinitely divisible and carries no intrinsic length; a jammed arrangement carries several—a particle scale, a contact-network correlation length, a core scale set by how many strong-rotation cells pack

together. The central technical content of the four pillars is that these arrangement lengths are exactly the lengths that the field description appears to “select” or “choose.” That coincidence, made precise and reproducible, is the evidence for Principle 1.

3 The marginal substrate: why a jammed arrangement flows

A jammed packing at the isostatic margin $z = 2d$ has zero shear reserve—the relaxed shear modulus vanishes, $G_{\text{relaxed}} \propto (z - 2d) \rightarrow 0$, while the bulk modulus stays finite—so it shears freely, resists compression, and carries one elastic speed $c_s^2 = B/\rho$: it *is* a fluid. We reproduce this locally (bootstrap $z_0 = 5.99$, CI $[5.49, 6.29] \ni 6$) and inherit the companion’s five-observable verification; rotation, costing a rigid packing nothing yet shearing a differential one, is the unjamming switch. The same two rules fix the laminar–turbulent transition’s universality class as directed percolation—confirmed by experiment (2016–2024) and by five exponents measured in this bundle.

A0 calls the medium “a coarse-grained, jammed configuration of finite-volume particles.” Before the axioms say how such a configuration *interacts*, there is a prior question the rest of the document silently leans on: why should a *jammed* configuration flow at all? Jammed packings are, after all, solids. This section answers it with the central inherited result of the companion whitepaper (*VP Theory*, concept DOI [10.5281/zenodo.17932566](https://doi.org/10.5281/zenodo.17932566)), reproduced locally, at reduced size, inside this bundle: a packing at the *isostatic margin* is the special solid whose shear rigidity is exactly zero while its compressive rigidity is not—which is the mechanical definition of a fluid. Everything in this section is dimensionless; nothing imports the companion’s dimensional anchor (Section 14).

3.1 Counting constraints: where $z = 2d$ comes from

Let a packing of N frictionless particles in d dimensions touch through N_c contacts; the mean coordination is $z = 2N_c/N$. Each contact is one scalar constraint (it fixes one inter-particle gap); each particle carries d degrees of freedom. For the packing to be rigid the constraints must pin the dN degrees of freedom (global rigid motions aside, an $\mathcal{O}(1)$ correction at large N):

$$N_c \geq dN - \frac{d(d+1)}{2} \quad \Longleftrightarrow \quad z \geq 2d - \mathcal{O}(1/N).$$

This is Maxwell’s counting argument, and it is the entire input. The point $z_{\text{iso}} = 2d$ is *isostatic*: exactly enough contacts to be rigid, with not one to spare—rigid the way a structure with zero redundancy is rigid; remove anything and it moves. The whole section lives on the distance to this margin,

$$\Delta z \equiv z - 2d \geq 0.$$

3.2 The fluidity theorem: zero shear reserve at the margin

Two moduli describe the linear mechanics of a packing. The *affine* (Born) moduli are what the contact network would resist if every particle followed the imposed strain homogeneously; the *relaxed* moduli let the particles re-equilibrate after the strain (the non-affine relaxation),

$$G_{\text{relaxed}} = G_{\text{Born}} - \frac{\Xi^T H \Xi}{V},$$

where H is the Hessian of the contact energy on the rigid backbone and $\Xi = -\partial^2 E / \partial \gamma \partial \mathbf{x}$ is the force mismatch the affine strain leaves behind. The relaxed modulus is what a rheometer measures.

[DERIVE] (inherited + locally reproduced) — the marginal packing is a fluid

As the margin is approached, the relaxed shear modulus vanishes linearly while the bulk modulus stays finite:

$$G_{\text{relaxed}} \propto \Delta z \rightarrow 0, \quad B \rightarrow \text{finite} \quad (\Delta z \rightarrow 0).$$

A medium with $G = 0$ and $B > 0$ deforms under arbitrarily small shear yet resists compression and transmits longitudinal sound: that is a fluid, by mechanical definition. *Fluidity is therefore not an assumption of A0; it is a property the marginal arrangement earns.*

Provenance. Verified in the companion bundle (jamming-spine verification v0.4, concept DOI 10.5281/zenodo.17932566) on five independent observables (table below), and reproduced locally at reduced N by `marginal_fluidity.py` (box below).

Failure mode. A correctly prepared, accepted ensemble whose bootstrap z_0 excludes $2d$ as N grows, or whose B collapses together with G , falsifies the statement (Section 14, row 0).

The companion’s verification is worth tabulating, because its strength is the *independence* of the observables that all point at the same margin:

Observable (companion)	Behaviour at the margin	Measured
Relaxed shear modulus	$G_{\text{relaxed}} \propto \Delta z \rightarrow 0$	$z_0 = 5.90/5.95$ ($N=512/1024$)
Soft-mode onset ω^*	vanishes at isostaticity	$z_0 = 6.015$, $R^2 = 0.997$
Viscosity proxy	diverges as $\Delta z \rightarrow 0$	confirmed
Quasistatic yield stress	$\sigma_y \rightarrow 0$	$z_0 = 5.98$
Athermal stress relaxation	residual $<1\%$	confirmed
Bulk modulus (the contrast)	stays $\mathcal{O}(1)$	0.90/1.13/1.40

Table 1: The companion’s five-observable verification of marginal fluidity (quoted, not re-derived; concept DOI 10.5281/zenodo.17932566). Five vanishing routes and one non-vanishing contrast agree on the same point $z = 2d$.

Reproducibility box: Local reproduction — marginal fluidity at reduced N (`marginal_fluidity.py`)

3D bidisperse harmonic spheres, $N \in \{128, 256\}$, 7 packing fractions $\times$ 4 seeds = 56 packings; Born moduli, mismatch vector and Hessian assembled analytically and self-verified against finite differences (Born 7.4×10^{-9} , Ξ 9.2×10^{-10} , H 3.4×10^{-10}); the non-affine term by a *pinned* positive-definite Cholesky solve with iterative refinement—never an eigendecomposition pseudoinverse (the soft-mode over-subtraction trap; algorithm box, Appendix B).

- **Fluidity intercept:** pooled fit $G_{\text{rel}} = a(z - z_0)$ gives $z_0 = 5.99$, bootstrap 95% CI $[5.49, 6.29] \ni 6 = 2d$ ($n = 56$); per size, $N=128$: 6.20 $[5.82, 6.42]$, $N=256$: 5.83 $[5.24, 6.15]$. **PASS**
- **Monotone:** $\langle G_{\text{rel}} \rangle$ over z -quartiles = 4.04, 6.74, 10.08, 12.39. **PASS**
- **Bulk survives:** lowest- z quartile $\langle B_{\text{rel}} \rangle = 47.0$ vs $\langle G_{\text{rel}} \rangle = 4.04$ —the shear reserve dies an order of magnitude ahead of the compressive one. **PASS**
- **Acoustic:** $c^2 = B_{\text{rel}}/\rho$ is size-independent to 0.4% (0.2887 vs 0.2875 in bundle units), softened below the affine Born value by the non-affine relaxation, $\langle B_{\text{rel}}/B_{\text{Born}} \rangle = 0.711$ (companion,

larger N : $\times 0.65$). **PASS**

- **Ledger:** 56/56 accepted (jammed, positive-semidefinite, residual force $< 10^{-7}$); negative G_{rel} kept-not-clamped: 0 occurrences here (2 kept in the companion’s larger runs). **PASS**

Slopes and moduli are in bundle units (they rescale with the contact stiffness); the intercept z_0 and the ratios are the invariant content.

3.3 Rotation is the unjamming switch

The margin does not only make the medium shearable; it singles out *rotation* as the operation that fluidizes it—the substrate-level origin of this document’s rotational theme.

Two elementary facts, both verified in `unjam_inflow.py`. (i) A *rigid* rotation $\mathbf{u} = \theta \hat{\mathbf{z}} \times \mathbf{r}$ changes no bond length at first order, identically: $\hat{\mathbf{n}} \cdot (\hat{\mathbf{z}} \times \mathbf{r}) = 0$ (measured $\leq 2.8 \times 10^{-17}$)—rotation per se costs the packing nothing. (ii) Any *differential* rotation is locally a shear, and the strain a shear can impose before breaking a contact is bounded by that contact’s overlap. Across the ensemble the overlap distribution collapses with pressure (quantile correlations $\text{corr}(\log q_{05}, \log P) = 0.96$, $\text{corr}(\log \text{med}, \log P) = 0.99$; densest-to-marginal ratio $5.9\times$): *the shear reserve is the overlap, and it vanishes at the margin*. Hence at $\Delta z \rightarrow 0$ any rotational shear, however small, exceeds the reserve, breaks contacts, and unjams the neighbourhood: *rotation is the fluidization switch*. The unjammed material then moves along the one direction the rotating demand leaves open—inward—which is the geometric origin of the directed-inflow nozzle ($\sim r^{-4}$) that Mechanism M and gate G-Q use downstream (Section 11); there it now enters as derived geometry, not as a postulate.

3.4 The acoustic face: one speed survives

At the margin the packing keeps exactly one stiffness, B , hence exactly one elastic wave—longitudinal sound at

$$c_s^2 = \frac{B}{\rho},$$

with no transverse counterpart ($G_{\text{relaxed}} = 0$): the defining acoustic signature of a fluid. The local measurement above puts numbers on it (c^2 size-independent to 0.4%; non-affine softening 0.711), and Pillar I derives the same speed from the balance laws by linearization (Section 6). What is *not* done here: the companion’s identification of the full-jamming elastic speed with the invariant speed of light rests on its dimensional anchor and is neither used nor claimed in this document (Section 14).

3.5 The constitutive ladder: where each closure lives

The margin organizes the constitutive zoo of continuum mechanics on a single axis. Reading Δz downward:

- **Deep in the jammed phase** (Δz large): finite G_{relaxed} and finite yield stress—an elastic/plastic solid.
- **Approaching the margin:** G_{relaxed} and σ_y vanish while the viscosity proxy grows—yield-stress and shear-thinning rheology; this is the regime where the Newtonian gate of Pillar I *fails* in exactly the long-memory, force-chain way it names.
- **At and just past the margin** ($\Delta z \rightarrow 0^+$): zero shear reserve, finite B —the Newtonian fluid is the leading small-Deborah, locally isotropic response of the *barely unjammed* medium. The Newtonian-closure gate of Pillar I is thereby *located* on this axis rather than free-floating (its content and **[GATE]** status are unchanged; its position is the new information).

The ladder is a reading of measured monotonicities (Table 1), not an additional theorem; each rung keeps the grade its evidence earns.

3.6 Why a driven medium sits at the margin (G-SOC)

One gate remains, and it is the hinge between this section and the dissipation pillar: *why should a flowing medium be at the margin at all times?*

[GATE] G-SOC: driven marginality (self-organized criticality)

Statement. Drive unjams (pushes z down toward $2d$); rest re-jams (pulls z up). A continuously driven medium is squeezed from both sides and *pins* at marginal coordination—self-organized criticality with $z = 2d$ as the critical manifold. If true, the substrate of any flowing medium is permanently marginal, and the event channel of Pillar IV is permanently available.

Status. The companion verification observes the pinning: under slow shear with re-jamming, the steady contact pressure parks at $g^*/g_0 \approx 1.1\text{--}1.3$ regardless of the starting point (concept DOI 10.5281/zenodo.17932566). What is demonstrated is pinning *near* the margin; what is *not* demonstrated—there or here—is arrival *exactly* at $z = 2d$ under drive. The companion flags this as its own open loop; we inherit the flag, not the conclusion.

Falsification route. Drive a marginal packing at a fixed slow rate with re-jamming and track the steady g^* (equivalently z) across drive amplitudes and seeds. *Systematic drift* of the steady value with drive or preparation—instead of pinning—falsifies self-organized marginality, and with it the identification of sustained flow with sustained criticality.

3.7 The transition test: a 140-year problem lands where the substrate says

The G-SOC rules are not decoration; they make a parameter-free, checkable statement about one of fluid mechanics’ oldest unsolved problems. The laminar–turbulent transition has resisted classification since Reynolds (1883). Its structure is peculiar and well documented: laminar pipe flow is linearly stable at every Reynolds number, so turbulence cannot arise spontaneously—it spreads only by contact from regions already turbulent—and it can locally die. “Drive unjams, rest re-jams,” read at the scale of turbulent patches, *is* this structure: an absorbing (laminar) state, local contagion, local recovery.

[DERIVE] — the substrate rules fix the universality class of the transition

A locally interacting system with a unique absorbing state and no additional symmetry has, generically, one continuous-transition universality class: directed percolation (DP). The G-SOC dynamics of §3.6 is such a system, so the framework fixes the class *in advance*, with no fluid input. This is the class the experiments found: quasi-1D Couette (2016, $\beta = 0.276$), channel flow (2016, four exponents and a scaling relation), and—closing the question—pipe flow itself (2024), whose authors name the phase above threshold a *jammed phase of puffs*. A hard problem, solved by others, lands exactly where the substrate says it must.

Local verification. The rules were transcribed literally to a lattice (pseudogap reserves $P(x) \propto x^\theta$, demand-exceeds-reserve unjamming, probabilistic re-jam) and five independent exponents were measured with no tuning (`transition_dp.py`, three stages; box below). All five land on DP; mean-field and compact-DP are excluded.

Falsification route. A subcritical shear-flow transition conclusively shown to lie outside DP breaks the category claim; within the lattice, any of the five gates failing on rerun

breaks the transcription.

Reproducibility box: Transition class — five exponents, no tuning
(transition_dp.py → refine → final)

1D ring; per-site reserve from the marginal pseudogap ($\theta=0.45$, flagged); inactive site unjams when demand $\lambda n_{\text{act}} U$ exceeds its reserve; active site re-jams with probability $q=0.35$; toggled sites redraw. $\lambda_c = 1.2544$ by curvature-minimised bisection.

exponent	measured	DP (1+1d)	
decay $\rho \sim t^{-\delta}$	0.158	0.1595	PASS
lifetime $\tau \sim \Delta^{-\nu_{\parallel}}$	1.637	1.7338	PASS
hyperscaling $\theta_s + \delta + \delta'$	0.662	$d/z = 0.6326$	PASS
spreading $N \sim t^{+\theta_s}$	0.335 (windows 0.382→0.335)	0.3137	PASS
activity corr. $M_2 - 1 \sim r^{-a}$	0.482 ± 0.011	$2\beta/\nu_{\perp} = 0.504$	PASS

The pseudogap exponent is a flagged parameter precisely because the class must not depend on it; that independence is the content of universality. (The SOC avalanche exponent at 0.35% below threshold has a cutoff exceeding 10^6 and is reported with thin statistics, not gated.)

One of these predictions is now closed and one boundary stated. (i) At transitional criticality the turbulent-fraction box roughness decays as $r^{-2\beta/\nu_{\perp}}$, an exponent that must track the spatial dimension of the DP class. Running the *same* G-SOC rules on a 2D lattice (4 neighbours $\Rightarrow$ (2+1)d DP) and measuring the box roughness in the critical quasi-stationary state gives the result below.

Reproducibility box: Transition roughness in quasi-2D — (2+1)d DP
(transition_dp.2d.py)

The 1D ring of the box above becomes an $L \times L$ lattice; the absorbing laminar state, the marginal-stability reserves ($\theta = 0.45$) and the re-jam $q = 0.35$ are unchanged — only the dimension changes, which is the universality statement. λ_c is located by the survival transition; in the critical state the box-roughness $M_2(r) - 1 = \langle \rho_r^2 \rangle / \langle \rho_r \rangle^2 - 1$ is a clean power law in r .

- **Quasi-2D roughness.** $2\beta/\nu_{\perp} = 1.475 \pm 0.058$ (6 realisations, $L = 384$), against the (2+1)d DP value $2\beta/\nu_{\perp} = 1.591$. **PASS**
- **Dimensional scaling captured.** With the quasi-1D measurement (0.482) the measured 2D/1D ratio is 3.06 against the DP ratio $1.591/0.504 = 3.16$; both dimensions sit $\sim 5\text{--}7\%$ low — a single, consistent finite-size bias, not a class mismatch.

$\Rightarrow$ the transition-critical roughness exponent tracks the DP spatial dimension exactly as predicted (quasi-1D 0.50, quasi-2D 1.59); the residual $\sim 5\text{--}7\%$ finite-size offset is the same **[GATE]** as for any finite lattice. The quasi-2D prediction — once flagged for Couette/channel reanalysis — is met internally by the model itself.

(ii) The superexponential puff lifetimes are the extreme-value (Gumbel) statistics of the weakest reserve inside a puff. And one boundary, stated plainly: the transition fixed point does *not* give the intermittency of developed turbulence ($\mu \approx 0.25$ against the 3+1d DP value ≈ 2.8). Developed flow is the *driven* phase beyond the transition; its exponents belong to the event RG and stand as that program's open problem.

3.8 What is inherited, what is local, what is not imported

For auditability, the boundary of this section in one place. *Inherited and locally reproduced:* the fluidity theorem and the acoustic ratio (Table 1 and the box above). *Inherited as derived*

geometry: rotational unjamming and the directed inflow (§3.3); the forced-radius attractor structure (Section 8). *Inherited as a graded gate*: G-SOC (§3.6). *Deliberately not imported*: the companion’s mass ladder, its α and δ derivations, its absolute (dimensional) anchors, and its speed-of-light identification. Citations use the concept DOI only; the companion’s grades translate by the rule stated in “How to read this document.”

4 The axiomatic core: rotation forces a three-body structure

A co-rotating pair cannot self-propel (No-Go), so the minimal interaction of a rotating arrangement is the irreducible three-body triangle U_3 . Energy leaves the conservative core only through rearrangement events, by $\varepsilon_{\text{bind}} = \Theta\sigma \geq 0$; both statements reproduce to 10^{-15} .

Before the four pillars, we lay the foundation they rest on, because it answers a question the field description never poses: *what is the minimal interaction a rotating arrangement must have?* The answer is sharp and surprising. A two-body law is not enough—it is provably unable to reproduce elementary fluid behaviour—and the deficit is repaired by an irreducible *three-body* term whose minimal unit is a triangle. This is the precise origin of the C_3 co-rotation of the companion physics whitepaper and of the mechanism of Section 11.4: the triangle is not a chosen motif but the smallest interaction a rotating medium admits. Section 3 already handed rotation its special status from below—a marginal arrangement cannot resist it—so the axioms of this section say what rotation must *interact through*, given that the substrate has already made it the operative motion.

4.1 State, invariants, and the two-body kernel

The arrangement of A0, specialized to rotation, is a set of cores $\{(\mathbf{x}_i, \mathbf{v}_i, L_i, \rho_i(\cdot))\}$: each core carries a position $\mathbf{x}_i$, a velocity $\mathbf{v}_i$, a spin (circulation) axial vector L_i , and an internal density profile ρ_i . We want the interaction energy of two such cores. Three physical demands constrain it almost completely. *Translation and Galilean invariance* forbid any dependence on absolute position or velocity, so the interaction may depend only on the relative position $\mathbf{r} = \mathbf{x}_i - \mathbf{x}_j$ and the spins. *Isotropy* (no preferred direction in the medium) means it must be a rotational scalar built from $\mathbf{r}$, L_i , L_j . *Finiteness* (a core has a smooth, axisymmetric profile, expanded only to second moment—the Ward axiom) keeps only terms up to quadratic in the spins.

There are exactly three independent scalars one can form at this order: the spin-independent 1, the spin–spin contraction $L_i \cdot L_j$, and the projection product $(L_i \cdot \hat{\mathbf{r}})(L_j \cdot \hat{\mathbf{r}})$. Each multiplies a radial function, and these functions are not independent—the same axisymmetric profile generates all three—so they collapse to derivatives of a *single source function* $F(r)$. The two-body kernel is therefore forced into

$$U_2(S_i, S_j) = A(r) + B(r)(L_i \cdot L_j) + C(r)(L_i \cdot \hat{\mathbf{r}})(L_j \cdot \hat{\mathbf{r}}), \quad \{A, B, C\} = (\text{derivatives of } F), \quad (1)$$

a one-degree-of-freedom family. This is not a modelling choice among many; given the three demands, it is the only quadratic-spin kernel there is. The force is $\mathbf{F}_i = -\nabla_{\mathbf{x}_i} U_2$, the torque follows from the spin gradient, and because U_2 depends on $\mathbf{x}_i$ only through $\mathbf{r}$, the pair obeys Newton’s third law $\mathbf{F}_i = -\mathbf{F}_j$ identically.

The Hamiltonian $H = K + \sum_{i < j} U_2$ then conserves energy, linear momentum $\mathbf{P} = \sum \mathbf{p}_i$, and angular momentum by construction: $\sum_i \mathbf{F}_i = 0$ gives $\dot{\mathbf{P}} = 0$. Which *quadratic* invariant accompanies the energy depends on dimension—in two dimensions it is the enstrophy, giving the pair (E, Z) ; in three dimensions it is the helicity $H = \int \mathbf{u} \cdot \boldsymbol{\omega}$, giving (E, H) . This duality is not decorative: it is precisely what fixes the *sign* of the cascade, inverse in 2D and forward in 3D (Pillar 9).

Algorithm: U2 is central and conserves momentum (axioms.py)

Input: two cores at positions X_0, X_1 (2D) with spins L_0, L_1 ; radial functions $A(r), B(r)$ (the 2D kernel; the C -term drops, see below).

Steps:

1. For each ordered pair compute $\mathbf{r} = X_i - X_j$, $r = |\mathbf{r}|$, $\hat{\mathbf{r}} = \mathbf{r}/r$.
2. Force $\mathbf{F}_i = -(A'(r) + B'(r)L_iL_j)\hat{\mathbf{r}}$ (note: along $\hat{\mathbf{r}}$, i.e. central).
3. Measure non-centrality $|\mathbf{F}_0 \times \hat{\mathbf{r}}|$ and the total force $|\mathbf{F}_0 + \mathbf{F}_1|$.

Output: non-centrality and $|\sum \mathbf{F}|$.

pass: non-centrality $\leq 10^{-16}$ (force is exactly central) and $|\sum \mathbf{F}| = 0$ (momentum conserved).
Measured: 2.8×10^{-17} and 0.

4.2 The No-Go theorem: two bodies are not enough

Proposition 1 (Two-body No-Go). *[DERIVE]. In two dimensions $L_i = L_i\hat{\mathbf{z}}$ and $\hat{\mathbf{r}} \perp \hat{\mathbf{z}}$, so $(L_i \cdot \hat{\mathbf{r}}) = 0$ and (1) collapses to a pure central force $U_2^{2D} = A(r) + B(r)L_iL_j$. A central-force system conserves total momentum, so its centre of mass cannot accelerate: a co-rotating pair has no self-propulsion. Yet a real 2D vortex dipole does self-propel, by its Biot–Savart (Kelvin) impulse—a degree of freedom no pairwise potential possesses. Moreover, no single F can simultaneously reproduce same-sign coaxial leapfrogging and opposite-sign stable separation in three dimensions. The two-body arrangement is therefore structurally incomplete.*

This is the rigorous content behind “a circle cannot be cleanly three-divided.” A medium of *paired* rotations cannot do what fluids visibly do; the missing physics is genuinely many-body.

The proof, step by step. The argument is short and worth seeing in full. (i) In two dimensions every spin points out of the plane, $L_i = L_i\hat{\mathbf{z}}$, while every separation lies in the plane, so $\hat{\mathbf{r}} \perp \hat{\mathbf{z}}$ and the projection $(L_i \cdot \hat{\mathbf{r}}) = 0$. (ii) The C -term of (1) therefore vanishes identically, leaving $U_2^{2D} = A(r) + B(r)L_iL_j$, a function of r alone—a *central* potential. (iii) A central pair force is $\mathbf{F}_i \parallel \hat{\mathbf{r}}$ and obeys $\mathbf{F}_i = -\mathbf{F}_j$, so $\sum_i \mathbf{F}_i = 0$ and the centre of mass satisfies $\ddot{\mathbf{X}}_{\text{cm}} = 0$: if it starts at rest it stays at rest. (iv) Hence no co-rotating pair governed by U_2 can translate itself. But a real opposite-sign dipole *does* translate—each vortex is carried in the Biot–Savart velocity of the other, at speed $\Gamma/2\pi d$ —and that velocity is a *field impulse* (the Kelvin impulse), not the gradient of any pair potential. The degree of freedom simply is not present in U_2 . The two-body arrangement is structurally incomplete, and (1) cannot be patched by any choice of F ; the repair must add a new term.

Algorithm: No-Go: two bodies cannot self-propel (axioms.py)

Input: two co-rotating cores ($L_0, L_1 > 0$) under the 2D kernel U_2 ; integration steps, dt .

Steps:

1. Record the initial centre of mass $X_{\text{cm}}^0 = \frac{1}{2}(X_0 + X_1)$.
2. Integrate $\ddot{X}_i = \mathbf{F}_i$ (forward Euler) under the central U_2 force for many steps.
3. Measure the CM displacement $|X_{\text{cm}} - X_{\text{cm}}^0|$.
4. Separately, evaluate the Biot–Savart translation speed of an opposite-sign dipole, $v = \Gamma/2\pi d$.

Output: CM displacement; dipole speed v .

pass: CM displacement $\leq 10^{-14}$ (no self-propulsion under U_2) while the real dipole has $v \neq 0$.
Measured: $\sim 10^{-15}$ and $v = 0.159$. The contrast *is* the No-Go.

4.3 The three-body term and its triangle

The minimal repair is a parity-even triple-product, the unique lowest-order invariant built from spins and relative positions,

$$U_3 = \sum_{i < j < k} D(P_{ijk}) S_{ijk}, \quad S_{ijk} = \sum_{\text{cyc}} L_i \cdot (\mathbf{r}_{ij} \times \mathbf{r}_{ik}), \quad D(P) = d_0 e^{-P/\ell}, \quad (2)$$

with P_{ijk} the triangle perimeter (a Yukawa-type screening, finite at contact, absolutely convergent at long range). It is translation/rotation invariant, exchange symmetric, parity-even, and Hamiltonian-admissible. Its two-dimensional reduction is the key:

$$\mathbf{r}_{ij} \times \mathbf{r}_{ik} = 2A_\Delta \hat{\mathbf{z}} \Rightarrow S_{ijk}^{2D} = 2A_\Delta (L_i + L_j + L_k), \quad (3)$$

where A_Δ is the *signed area* of the triangle (i, j, k) . Because A_Δ depends on all three vertices, the third core acts as a *mediator* that induces a transverse force on the (i, j) pair—exactly the self-propulsion channel the No-Go theorem forbids to two bodies. The smallest arrangement that behaves like a fluid is a *triangle*, and the quantity that activates it is the signed triangle area coupled to the spin sum.

Remark (Why this is the C_3 of the lattice, and the co-rotation). The lattice realization of this triangle is the C_3 unit of the $82 = 81 + 1$ core: the three-fold ring is the discrete triple, and its single unsatisfiable contact—the residual of trying to three-divide a circle—is the central nozzle (Section 11.4, G-Q). The frustration that forces co-rotation (Mechanism M, Step 1) is the discrete shadow of (3): a triangle of spins cannot cancel, so its signed area drives a net transverse flow. The continuum triple-product and the lattice frustration are the same statement.

Algorithm: U3 reduces to signed area $\times$ spin sum (axioms.py)

Input: three cores at X_0, X_1, X_2 (2D) with spins L_0, L_1, L_2 .

Steps:

1. Direct triple-product: $S = \sum_{\text{cyc}} L_i [(\mathbf{r}_{ij} \times \mathbf{r}_{ik})_z]$, using $(\mathbf{a} \times \mathbf{b})_z = a_x b_y - a_y b_x$.
2. Signed triangle area $A_\Delta = \frac{1}{2} (\mathbf{r}_{01} \times \mathbf{r}_{02})_z$.
3. Compare S with $2A_\Delta(L_0 + L_1 + L_2)$ over random triangles.
4. Build $U_3 = d_0 e^{-P/\ell} S$ (P the perimeter); translate the $(0, 1)$ pair CM and take $\mathbf{F}_{\text{cm}} = -\partial U_3 / \partial X_{\text{cm}}$ by finite difference; measure the component of $\mathbf{F}_{\text{cm}}$ transverse to $\mathbf{r}_{01}$.

Output: reduction error $|S - 2A_\Delta \sum L|$; transverse force.

pass: reduction error $\leq 10^{-15}$ (the identity holds to machine precision); transverse force $\neq 0$ (measured $|\mathbf{F}| = 0.145$, transverse = 0.139)—the channel the No-Go forbids to pairs is restored by the triangle.

4.4 Event dissipation: how the arrangement sheds energy

The Hamiltonian core (1)–(2) conserves energy; dissipation enters through a separate, symmetric (metriplectic) channel tied to *rearrangement events*—mergers and collapses that release binding energy. The governing law is

$$\varepsilon_{\text{bind}} = \Theta \sigma, \quad \sigma \geq 0, \quad \left. \frac{dH}{dt} \right|_{\mathcal{J}} = -\varepsilon_{\text{bind}} \leq 0, \quad \frac{dS}{dt} \geq 0, \quad (4)$$

with the metriplectic compatibility $G\nabla C = 0$ (Casimirs preserved, entropy monotone). Equation (4) is the precise form of the event measure of Pillar IV: it is what lets dissipation survive the inviscid limit, because the binding release $\varepsilon_{\text{bind}}$ does not vanish with the molecular viscosity.

4.5 What is proved, and what this whitepaper verifies

The axiomatic core is supported by a suite of theorems—uniqueness of the kernel, the No-Go, the constructive sufficiency of U_3 , non-blowup of the jump dynamics, an H -theorem, the inviscid-dissipation identity, flux-sign invariance, an RG fixed point, metriplectic consistency, identifiability, cluster bounds, Γ -convergence to the continuum, and a large-deviation principle for coherent structures. The four pillars of this whitepaper are the *independently reproduced* core of that suite; the correspondence is exact.

Axiomatic theorem	What it states	Verified here by
Kernel uniqueness / $U_2=F(r)$	two-body law is one source function	Pillar I (balance laws)
No-Go (two-body)	no 2D self-propulsion from pairs	Prop. 1; Mechanism M
U_3 sufficiency (three-body)	triangle restores fluid behaviour	Mechanism M; G-Q test
Inviscid dissipation	$\varepsilon_{\text{total}} = \varepsilon_{\text{bind}} > 0$ as $\nu \rightarrow 0$	Pillar IV; U1 (Burgers)
Flux-sign invariance	2D inverse / 3D forward (E, Z vs E, H)	Pillar IV; E3 (2D/3D)
Metriplectic consistency	$G\nabla C = 0, dS/dt \geq 0$	Pillar IV (metriplectic)
Γ -convergence ($a \rightarrow 0$)	core lattice $\rightarrow$ continuum energy	Pillar I (continuum limit)
Length selection / Π -line	selected scale, 1/2 exponent	Pillar III; U3 (Turing)

Table 2: The axiomatic theorems and the reproductions in this whitepaper. The pillars are not separate claims; they are the verified core of the axiomatic vortex dynamics, read through “form follows arrangement.”

The core, in one line. A rotating arrangement cannot be built from pairs (No-Go); its minimal interaction is a three-body triangle whose signed area drives the flow ((2)); and it sheds energy through rearrangement events ($\varepsilon_{\text{bind}} = \Theta\sigma$) that survive the inviscid limit. The four pillars verify this core; the rest of the document develops it.

5 Dimensionless diagnosis: the Π -invariants and the event RG

A selected arrangement is diagnosed by three invariants (Π_T, Π_L, Π_{ST}) and explained by an event RG whose fixed point, $(1 + \alpha_\star) = \psi + \chi$, is where Π_L stops flowing. The one-half law of Pillar III is this plateau read in real space.

The axiomatic core says what the interaction is; the present section says how to *recognize* a selected arrangement and how it flows under coarse-graining. Both are dimensionless statements, and both make the thesis operational: the form a flow settles into is the fixed point of an event-driven renormalization, and one can read off from data whether a given structure sits at that fixed point.

5.1 The Π -invariants

A quasi-stationary arrangement at “optimal contact”—the configuration the medium relaxes into and holds—is diagnosed by three dimensionless invariants: a temporal Π_T , a spatial Π_L , and a spatio-temporal coupling Π_{ST} . At an optimal arrangement the first two are pinned, $\Pi_T \approx 1$ and $\Pi_L \approx 1$, while the coupling invariant carries the criterion. The operational test (ST1') is a *line*: in the log plane the coupling Π_{ST} holds a straight line of slope $-a/b$ along an optimal trajectory (transitions between arrangements included), and the orthogonal departure

from that line is a single defect, the departure of Π_{ST} from that line:

$$\delta_{ST} = (\text{departure of } \Pi_{ST} \text{ from the line of slope } -a/b), \quad (5)$$

and the arrangement is judged optimal when $\Pi_T \approx 1$, $\Pi_L \approx 1$, and $\delta_{ST} \leq 0.05$. The defect is the trigger of the document’s reproducibility discipline (a fail-fast rule): a structure with δ_{ST} above tolerance is flagged for splitting, deferral, or exclusion rather than reported as a clean selection. This is what makes “form follows arrangement” falsifiable in practice—an arrangement either holds the line or it does not.

5.2 The event renormalization group

Under coarse-graining by a factor λ , the event-dissipation law (4) flows together with the selected length: the binding release weakens as $\varepsilon_{\text{bind}} \mapsto \lambda^{-\chi} \varepsilon_{\text{bind}}$, the effective stiffness as $\sigma_{\text{eff}} \mapsto \lambda^\psi \sigma_{\text{eff}}$, the length as $L \mapsto \lambda L$, and the cascade exponent α may run. The spatial invariant

$$\Pi_L = \frac{L^{1+\alpha} \varepsilon_{\text{bind}}}{\sigma_{\text{eff}}} \implies \frac{d \log \Pi_L}{d \log \lambda} = (1 + \alpha) - \psi - \chi \quad (6)$$

therefore stops flowing exactly at the fixed point. The selected arrangement *is* that fixed point:

$$(1 + \alpha_*) - \psi - \chi = 0 \iff \Pi_L \text{ plateau} \quad (7)$$

and the sign of $(1 + \alpha) - \psi - \chi$ gauges whether a structure is strengthening (> 0) or weakening (< 0) as it is coarse-grained—the same monotonicity that the special-edition typhoon protocol uses to call intensification versus dissipation.

Reproducibility box: Event-RG fixed point = Π_L plateau (event_rg.py)

Iterating the recursion $L \mapsto \lambda L$, $\varepsilon_{\text{bind}} \mapsto \lambda^{-\chi} \varepsilon_{\text{bind}}$, $\sigma_{\text{eff}} \mapsto \lambda^\psi \sigma_{\text{eff}}$ and tracking Π_L :

- $(1 + \alpha) - \psi - \chi = 0$: Π_L is exactly scale-invariant ($1.000 \rightarrow 1.000$ over 12 steps)—a *plateau*. **PASS**
- > 0 (e.g. $+0.7$): Π_L grows ($1.0 \rightarrow 7.5$); < 0 (e.g. -0.6): Π_L decays ($1.0 \rightarrow 0.18$). **PASS**

The 1/2 length law of Pillar 8 is this plateau read off in real space; the JFM exponent is the fixed-point exponent, and δ_{ST} is the measured departure from it.

So the layers close on each other: the substrate (§3) makes the medium a fluid and hands rotation the unjamming switch, the axioms (§4) fix the interaction, the Π -invariants diagnose when an arrangement is selected, and the event RG explains *why* that arrangement is the one selected—it is the only scale-free configuration of the binding-release and stiffness flow. (Section 3 names the physical candidate for *how* a driven medium reaches and holds that fixed point: self-organized marginality, gate G-SOC.) The four pillars now follow as the in-domain verification of this structure.

6 Pillar I — Structural: arrangement forces the equations

From mass and pairwise interaction, $\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$ and $\rho D\mathbf{u}/Dt = \nabla \cdot \sigma$ (registry $T \equiv \sigma$) are exact identities; the Newtonian closure is a located, falsifiable gate. Solver corollaries hold to 3×10^{-8} and 4×10^{-6} , and the compressible corollary carries the substrate sound speed $c_s^2 = B/\rho$.

The first pillar establishes the weakest, most secure claim, and the one on which the others lean: *the form of the fluid equations is not postulated; it is forced by the arrangement*. We do not need to assume that the medium “obeys Navier–Stokes.” We need only A0. The balance laws then follow as exact identities, and the Newtonian closure follows conditionally.

6.1 From arrangement to fields

Given the kernel W_ℓ (smooth, normalized, $\int W_\ell = 1$), define

$$\rho(\mathbf{x}, t) = \sum_i m_i W_\ell(\mathbf{x} - \mathbf{x}_i), \quad \rho \mathbf{u}(\mathbf{x}, t) = \sum_i m_i \mathbf{v}_i W_\ell(\mathbf{x} - \mathbf{x}_i). \quad (8)$$

These are definitions ([**LOCK**]): the fields are weighted bookkeeping of the arrangement. No dynamical assumption has entered.

6.2 Mass and momentum balance are exact

Proposition 2 (Exact balance laws). [**DERIVE**]. *If the particles move by $\dot{\mathbf{x}}_i = \mathbf{v}_i$ and exchange momentum through pairwise forces, then (8) satisfy, identically,*

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (9)$$

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = \nabla \cdot \sigma + \rho \mathbf{f}, \quad \sigma = -p \mathbb{I} + \tau. \quad (10)$$

The stress σ is given exactly by an Irving–Kirkwood formula: a kinetic part from peculiar-velocity transport and an interaction part from pairwise forces integrated along the segment joining each pair.

The derivation, in full. Both identities come from differentiating the definitions (8) and using nothing but $\dot{\mathbf{x}}_i = \mathbf{v}_i$ and Newton’s third law. For continuity, differentiate ρ in time and use $\partial_t W_\ell(\mathbf{x} - \mathbf{x}_i) = -\dot{\mathbf{x}}_i \cdot \nabla_{\mathbf{x}} W_\ell = -\mathbf{v}_i \cdot \nabla_{\mathbf{x}} W_\ell$:

$$\partial_t \rho = \sum_i m_i \partial_t W_\ell(\mathbf{x} - \mathbf{x}_i) = -\nabla_{\mathbf{x}} \cdot \sum_i m_i \mathbf{v}_i W_\ell(\mathbf{x} - \mathbf{x}_i) = -\nabla \cdot (\rho \mathbf{u}), \quad (11)$$

which is (9), exactly, with no closure. For momentum, differentiate $\rho \mathbf{u}$:

$$\partial_t(\rho \mathbf{u}) = \sum_i m_i \dot{\mathbf{v}}_i W_\ell - \nabla_{\mathbf{x}} \cdot \sum_i m_i \mathbf{v}_i \otimes \mathbf{v}_i W_\ell. \quad (12)$$

The second sum is split by writing each particle velocity as the local mean plus a peculiar part, $\mathbf{v}_i = \mathbf{u}(\mathbf{x}_i) + \mathbf{c}_i$; the mean part assembles the convective flux $\rho \mathbf{u} \otimes \mathbf{u}$ while the fluctuations leave a *kinetic stress* $\sigma^{\text{kin}} = -\sum_i m_i \mathbf{c}_i \otimes \mathbf{c}_i W_\ell$. In the first sum $m_i \dot{\mathbf{v}}_i = \mathbf{F}_i = \sum_{j \neq i} \mathbf{F}_{ij} + m_i \mathbf{f}$. The pairwise piece is converted to a divergence by the Irving–Kirkwood identity: since $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$, pairing terms and writing the difference of two kernels as the integral of a gradient along the bond $\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{x}_j$ gives

$$\sum_i \left(\sum_{j \neq i} \mathbf{F}_{ij} \right) W_\ell(\mathbf{x} - \mathbf{x}_i) = \nabla_{\mathbf{x}} \cdot \sigma^{\text{int}}, \quad \sigma^{\text{int}} = -\frac{1}{2} \sum_{i \neq j} \mathbf{F}_{ij} \otimes \mathbf{r}_{ij} \int_0^1 W_\ell(\mathbf{x} - \mathbf{x}_j - s \mathbf{r}_{ij}) ds. \quad (13)$$

Collecting terms yields (10) with $\sigma = \sigma^{\text{kin}} + \sigma^{\text{int}}$ and $\rho \mathbf{f} = \sum_i m_i \mathbf{f} W_\ell$. Every step is an identity: the continuum balance laws are not a model of the arrangement; they *are* the arrangement, re-summed. The pressure is the isotropic part of σ and the deviatoric remainder is τ ; the kinetic part dominates in dilute, fast media and the interaction part in dense, jammed ones, but the *form* (10) is fixed regardless.

Why this matters for the thesis. The continuity and momentum *forms* in (9)–(10) are consequences of three facts about the arrangement: there is mass, it moves, and it pushes on itself. There is no separate axiom “the medium is a fluid”—and Section 3 showed none is needed: the marginal arrangement *earns* fluidity as a measured property. In this sense the *shape* of fluid dynamics is already configured: it is what the bookkeeping of any finite-mass, interacting arrangement must look like. The only genuinely open object is the stress σ , and the rest of the pillar is about when σ takes the Newtonian form. (For cross-document searchability we also record the registry form of the two laws, $\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$ and $\rho D\mathbf{u}/Dt = \nabla \cdot T$ with $T \equiv \sigma$; the symbol σ is used throughout this document.)

6.3 The Newtonian closure is conditional, not automatic

Closure as a gate, not an assumption

[GATE] (G-Newton). The interaction stress reduces to the Newtonian form

$$\tau = 2\mu S + \lambda(\nabla \cdot \mathbf{u})\mathbb{I}, \quad S = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top) - \frac{1}{3}(\nabla \cdot \mathbf{u})\mathbb{I}, \quad (14)$$

if and only if the arrangement's stress responds to deformation by local, isotropic, Markovian linear response in the declared regime. This requires three measurable conditions to pass: (i) stress–strain-rate linearity; (ii) effective isotropy after window averaging; (iii) a short memory (small Deborah number, $\text{De} \ll 1$). When they pass, (9)–(10) become the compressible Navier–Stokes equations, and in the incompressible limit

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho_0} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad \nu = \mu / \rho_0. \quad (15)$$

Failure mode. If linearity or short memory fails (jammed force chains with long memory, strong anisotropy that survives averaging), the closure is non-Newtonian and (14) is simply wrong for that regime. The gate is therefore falsifiable per regime.

Where the gate sits (Section 3). The three conditions are not free-floating: they name a position on the distance-to-margin axis $\Delta z = z - 2d$ of the substrate. Far from the margin, rearrangements are fast and local (short memory, small Deborah number) and the closure holds; as $\Delta z \rightarrow 0$ the relaxed shear stiffness and the yield scale vanish while the viscosity proxy diverges (inherited verification, Section 3), and the closure fails in exactly the jammed, force-chain way named above. The gate is thereby *located*, not merely stated.

The viscosity inherits a definite sign from the same construction: because the dissipative part of the stress is a positive quadratic form in the strain rate (a Green–Kubo integral of a stress autocorrelation, which is nonnegative), one has $\mu \geq 0$ and hence $\varepsilon_\nu \geq 0$ (**[DERIVE]**). The *value* of μ in a concrete arrangement is a gate item (it must be measured by Green–Kubo or dissipation matching), but its *sign* is forced.

6.4 What the solver confirms

Two corollaries of Pillar I are exact and therefore make sharp numerical targets. First, with no viscosity and no forcing the incompressible flow is a Hamiltonian (Lie–Poisson) system and conserves both energy and enstrophy in two dimensions. Second, with viscosity but no forcing the energy obeys the exact budget

$$\frac{dE}{dt} = -\varepsilon_\nu = -2\nu Z, \quad E = \frac{1}{2}\langle |\mathbf{u}|^2 \rangle, \quad Z = \frac{1}{2}\langle \omega^2 \rangle, \quad (16)$$

which is the bookkeeping backbone of Pillar IV. We verify both with a self-contained pseudo-spectral solver.

Reproducibility box: Pillar I — exact corollaries of the balance laws (ns2d.py, validate_all.py)

2D incompressible Navier–Stokes in vorticity form, periodic box $L = 2\pi$, integrating-factor RK4, 2/3 dealiasing; band-limited random initial vorticity, fixed seed.

- **Inviscid invariants** ($\nu = 0$, $N = 128$, $T = 4$): relative drift $\Delta E/E_0 = 2.9 \times 10^{-8}$, $\Delta Z/Z_0 = 5.4 \times 10^{-6}$. **PASS**
 $\Rightarrow$ confirms the spectral discretization and dealiasing realize the Hamiltonian structure of Proposition 2 in the inviscid limit.

- **Energy-budget identity** (16) ($\nu = 0.01$, $N = 128$, $T = 6$): $\max_t |(-\dot{E}) - \varepsilon_\nu|/\varepsilon_\nu = 4.4 \times 10^{-6}$. **PASS**
 $\Rightarrow$ the exact budget that Pillar IV will split into viscous and event channels holds to machine-gradient precision.
- **Enstrophy monotone (no forcing)**: $dZ/dt \leq 0$ at every step, so $\max_t Z = Z(0)$ and hence $\max_t \varepsilon_\nu = 2\nu Z(0) \propto \nu$ exactly. **PASS**

Run `python validate_all.py`; checks 1–3 correspond to the three bullets.

6.5 The compressible corollary: the arrangement has a sound speed

The balances (9)–(10) are not restricted to the incompressible regime in which the solver above runs. Linearize them about rest ($\rho = \rho_0 + \delta\rho$, $\mathbf{u}$ small) with a barotropic closure $p = p(\rho)$: continuity gives $\partial_t \delta\rho = -\rho_0 \nabla \cdot \mathbf{u}$, momentum gives $\rho_0 \partial_t \mathbf{u} = -c_s^2 \nabla \delta\rho$, and together

$$\partial_t^2 \delta\rho = c_s^2 \nabla^2 \delta\rho, \quad c_s^2 = \frac{dp}{d\rho} = \frac{B}{\rho},$$

with B the bulk modulus. This is textbook acoustics; what the arrangement adds is that B is not a free parameter but a *measured* property of the packing, and that at the margin it is the *only* stiffness left: $G_{\text{relaxed}} \rightarrow 0$ while B stays finite (Section 3), so the marginal medium supports exactly one elastic wave—longitudinal sound, the defining acoustic signature of a fluid. The substrate measurement puts numbers on it: $c^2 = B_{\text{rel}}/\rho$ agrees across system sizes to 0.4%, softened below the affine (Born) prediction by the non-affine relaxation ($\langle B_{\text{rel}}/B_{\text{Born}} \rangle = 0.711$ here; $\times 0.65$ in the companion’s larger- N runs). Scope, declared honestly: the pseudo-spectral solver of this pillar is incompressible, so the acoustic face is verified at substrate level (`marginal_fluidity.py`), not by a compressible flow run ([**DERIVE**] for the corollary given measured B ; the companion’s speed-of-light identification is *not* imported—see Section 14).

Pillar I, in one line. The equations of fluid motion are the necessary bookkeeping of a massive, interacting arrangement ([**DERIVE**]); their Newtonian specialization is a falsifiable regime property ([**GATE**]); their inviscid invariants and energy budget reproduce to 10^{-6} – 10^{-8} .

7 Pillar II — Geometric: arrangement fixes length

The rotational core capacity identity $N_\star r_{\text{eff}}^2 = \phi s L_{\text{core}}^2$ fixes the effective core length exactly once the arrangement is specified. The published cross-regime audit reproduces to 3×10^{-12} , the identity itself to 10^{-15} .

Pillar I gives the equations. Pillar II makes the first sharp statement about *form*: once the arrangement of a rotational core is specified, its effective length is *fixed* by an accounting identity. We then audit, across four flow regimes, exactly when a single length is an adequate summary of the arrangement and when it is not—turning the question “does this flow have a core radius?” into a measurable, reproducible diagnostic.

7.1 The rotational core capacity identity (RCCI)

Let an Okubo–Weiss–type criterion define a core mask C of area A_{core} . Let $\{B_k\}_{k=1}^{N_\star}$ be the strong-rotation blobs inside C , with areas A_k and equivalent radii $r_k = \sqrt{A_k/\pi}$. Define the RMS-equivalent radius, the filling fraction, and the shape factor by

$$r_{\text{eff}}^2 = \frac{1}{N_\star} \frac{1}{\pi} \sum_{k=1}^{N_\star} A_k, \quad \phi = \frac{\sum_k A_k}{A_{\text{core}}}, \quad s = \frac{A_{\text{core}}}{\pi L_{\text{core}}^2}. \quad (17)$$

Proposition 3 (RCCI — arrangement fixes the effective length). [\[LOCK\]](#)/[\[DERIVE\]](#). By construction $\sum_k A_k = \phi A_{\text{core}}$, hence

$$\boxed{N_\star r_{\text{eff}}^2 = \phi s L_{\text{core}}^2} \quad (18)$$

and, solving for the effective length,

$$r_{\text{pred}} = L_{\text{core}} \sqrt{\frac{\phi s}{N_\star}}. \quad (19)$$

Equation (18) is an exact accounting equality for a given mask and blob partition.

Remark (The same identity in any dimension). Written in terms of volumes rather than areas, (18) is dimension-free: $N_\star v_{\text{eff}} = \phi V_{\text{core}}$, where v_{eff} is the effective cell volume and V_{core} the core volume in d dimensions. This is the form that recurs across the document—it is the capacity identity verified in $d = 2, 3, 4, 5$ in Section 10 (U2), and it is *literally* the genome-condensate identity $N v_{\text{eff}} \simeq \phi V_{\text{core}}$ of Section 11 (E1). The geometric pillar is therefore not a two-dimensional accident; it is a single packing law, instantiated here in two dimensions. The fill fraction ϕ that enters the identity is, moreover, the same control knob whose marginal value the companion bundle *measures* for the substrate ($\phi_{\text{jam}} \approx 0.633\text{--}0.64$ at $z \rightarrow 2d$; see E1 of Section 11 and the numeric-adjacency non-claim of Section 14).

Why this is the geometric face of the thesis. The four numbers $(L_{\text{core}}, \phi, s, N_\star)$ are a complete coarse description of the *arrangement* of strong rotation inside the core: a length, how full the core is, how non-circular it is, and how many sub-cores pack into it. The identity says that the radius an independent observer would measure is not free: it is fixed by the arrangement. The single number r_{eff} is the arrangement, compressed. When a real measurement r_{meas} departs from r_{pred} , the departure is a quantitative statement that the flow *resists* compression to one length—i.e. that its arrangement is multiscale.

7.2 The cross-regime audit and what it reveals

Embedding (19) in four regimes turns the identity into a test of when “one core length” is a faithful summary. The arrangement is supplied by an Okubo–Weiss pipeline (the “A-pipeline”), while r_{meas} is measured by three independent segmenters (swirling strength λ_{ci} , a local-circulation Γ_2 proxy, and a persistence-based method). The normalized residual

$$\text{resid}_i = 100 \frac{|N_\star r_{\text{meas},i}^2 - \phi s L_{\text{core}}^2|}{\phi s L_{\text{core}}^2} \quad (20)$$

measures geometric breakdown sample by sample.

The qualitative picture (respecting the phenomena). In coherent single-core regimes (tropical cyclone, ocean eddy) the prediction tracks the measurement to a few percent and the residuals stay small: a one-length description is adequate, so the arrangement is well summarized by a single scale. In dense two-dimensional turbulence the errors become order unity, the rank correlation between r_{pred} and r_{meas} turns *negative*, and the residuals develop heavy tails: no single core length is adequate, because the arrangement is genuinely multiscale (filaments and merging events). The multi-vortex surrogate sits in between, with a $\sqrt{N_\star}$ penalty that is visible and method-dependent. This ordering—coherent $\rightarrow$ multi-core $\rightarrow$ filamented—is exactly the ordering a configurational view predicts: the more the arrangement refuses to collapse to one cell, the more the one-length summary fails.

The single most telling entry is the *negative* rank correlation in two-dimensional turbulence. It is not noise: it concentrates in frames with elongated, strain-dominated filaments and vortex mergers, where the Okubo–Weiss mask fuses several interacting vortices. In other words,

Regime	L_{core}	$N_{\star}$	rel. med.	$C_{10\%}/C_{20\%}$	Spearman ρ
Tropical cyclone	12.6	1	4.1%	93%/100%	+0.80
Ocean eddy	110	1	6.9%	71%/98%	+0.51
TC multi-vortex	12.8	2	18%	29%/56%	-0.08
2D turbulence	0.93	1	55%	10%/19%	-0.26

Table 3: The cross-regime audit, swirling-strength (λ_{ci}) segmenter (the strongest of the three; the shear-sensitive Γ_2 proxy is uniformly worse and is retained as a stress test). Median relative error of r_{pred} versus the measured radius, coverage within $\pm 10\% / \pm 20\%$, and the $r_{\text{pred}} - r_{\text{meas}}$ rank correlation. The ordering coherent (4–7%, $\rho > 0$) $\rightarrow$ multi-core (18%) $\rightarrow$ filamented (55%, $\rho < 0$) is the audit’s robust conclusion. All reproduced by `verify_rotcore.py`.

the geometric breakdown of the one-radius description is the *spatial signature of the enstrophy cascade*—the same fragmentation of coherent cores into filaments that, in Pillar 9, carries the forward enstrophy flux onto thin vorticity-gradient events. The geometric pillar and the dissipative pillar are thus reading one physical process from two angles: when the arrangement stops being a single cell (Pillar II’s residual blows up), it is precisely because it is cascading enstrophy to small scales through filamentary events (Pillar 9’s flux concentration). That two independent diagnostics locate the same transition is a coherence check on “form follows arrangement.”

Reproducibility box: Pillar II — the geometric identity and its audit (`verify_rotcore.py`)

Recomputation from the sample-level table (4800 samples; 12 case $\times$ method groups of 400) against the published audit tables.

- **Identity check.** $r_{\text{pred}} = L_{\text{core}} \sqrt{\phi s / N_{\star}}$ holds to a maximum relative error of 9.9×10^{-16} across all 4800 samples. **PASS** (Equation (19) is satisfied to floating-point.)
- **Robust metrics.** Median relative error, MAPE, MAE, NRMSE and Spearman ρ for every group reproduce the published table to a worst-case relative mismatch of 2.8×10^{-12} . **PASS**
- **Identity residuals.** Median/mean of (20) per group reproduce the published residual table (e.g. 2D turbulence: median ≈ 85 –91%, mean ≈ 127 –141%; coherent cores: single to low tens of percent). **PASS**

Run `python verify_rotcore.py <data_dir>` on the frozen sample table.

Reading the residuals as a configuration diagnostic. The gap between the median and the mean residual is a measure of how heavy the tail is—i.e. how often the arrangement produces a frame in which a single length is badly wrong. Small gap (coherent cores) means the arrangement is reliably single-cell; large gap (2D turbulence under the shear-sensitive Γ_2 proxy) means rare but extreme multiscale events. This is the sense in which the residual is not a “model error” so much as a *quantitative complexity parameter of the arrangement*.

Remark (What Pillar II does *not* claim). The RCCI is a kinematic sum rule, not a dynamical law: it does not predict how a core evolves, and it introduces no constitutive relation. Its content is exactly that the effective length is fixed by the arrangement, and that departures are measurable. The audit’s conclusions are stated as a regime/method *ordering*, which we expect to be robust to reasonable changes of error metric, rather than as absolute percentages.

Pillar II, in one line. Once the rotational arrangement ($L_{\text{core}}, \phi, s, N_{\star}$) is fixed, the effective core length is fixed by an exact identity (**[DERIVE]**), reproduced to 10^{-15} ; the cross-regime audit (reproduced to 10^{-12}) measures exactly when one length is enough and when the arrangement is irreducibly multiscale.

8 Pillar III — Dynamical: arrangement selects length

The dispersion $\lambda(k) = \mu + \varepsilon k^2 - \sigma k^4$ selects $L_\star = 2\pi\sqrt{2\sigma/\varepsilon} \propto (\sigma/\varepsilon)^{1/2}$, exactly one-half and μ -independent; nonlinear and blind DNS recover $\hat{\beta} \approx 0.501$. The selected radius is moreover a *global attractor* of the inherited stiffness-versus-inflow balance, with $F'(x^\star) = -(\pi/2)^5$.

Pillar II fixes a length *given* an arrangement. Pillar III is stronger and more surprising: it shows how a length is *selected* in the first place. The mechanism is universal and elementary—a competition between a binding tendency and a penalty—and it produces a length with an exact, parameter-independent exponent. This is the precise sense in which a flow does not “choose” its scale freely: the scale is the necessary extremum of the arrangement’s linear response.

8.1 Selection as an extremum, not a cascade

Consider the linear growth rate of perturbations at wavenumber k ,

$$\lambda(k) = \mu + \varepsilon k^2 - \sigma k^4, \quad \varepsilon, \sigma > 0. \quad (21)$$

Here μ is a uniform growth offset, the $+\varepsilon k^2$ term is a *binding* tendency that favors finer structure, and the $-\sigma k^4$ term is a *penalty* that suppresses it. The two compete, and the competition has a unique interior maximizer.

Proposition 4 (One-half length selection). *[DERIVE]. Stationarity $\partial_k \lambda = 2\varepsilon k - 4\sigma k^3 = 0$ selects*

$$k_\star = \sqrt{\frac{\varepsilon}{2\sigma}}, \quad L_\star = \frac{2\pi}{k_\star} = 2\pi\sqrt{\frac{2\sigma}{\varepsilon}} \propto \left(\frac{\sigma}{\varepsilon}\right)^{1/2}, \quad (22)$$

independently of the growth offset μ . Consequently the control exponent is exactly

$$\frac{d \log L_\star}{d \log(\sigma/\varepsilon)} = \frac{1}{2}. \quad (23)$$

Why μ -independence is the point. The offset μ controls *whether* the pattern grows; it does not touch *which* scale is selected. Selection is therefore a property of the *shape* of the response (the balance of binding and penalty), not of its overall strength. In the configurational reading, ε and σ are set by how the arrangement binds and how it resists curvature; the selected length is set by the square-root balance of the two, and nothing else. The exponent $1/2$ is not fit—it is the exponent of a square root.

Remark (The $1/2$ law is the real-space face of the RG fixed point). Pillar III and the event RG of Section 5 are the same statement seen twice. The selected $L_\star$ is exactly the scale at which Π_L stops flowing—the plateau of (7)—and the fixed-point exponent read off in real space is the one-half of (23). The shape balance εk^2 versus σk^4 here, and the flow balance $(1 + \alpha_\star) = \psi + \chi$ there, are two descriptions of the single fact that a configured medium has one scale-free length. The defect δ_{ST} of Section 5 is, correspondingly, the measured departure from the one-half line.

8.2 Selection is an attractor, not only an extremum

The extremum of (21) says *which* length wins the linear race. The inherited substrate adds a stronger, dynamical statement: the selected radius is a *global attractor* of the stiffness-versus-inflow balance. Write the competition in its normalized one-dimensional form,

$$\dot{x} = \alpha x^{-5} - x^{-4}, \quad x > 0,$$

where x is the core radius in selected units, x^{-4} is the directed-inflow term whose geometric origin is the rotational unjamming of the margin (Section 3), and αx^{-5} is the stiffness term. The

constant $\alpha = 2/\pi$ enters *as an inherited constant of the companion* (concept DOI 10.5281/zenodo.17932566); its derivation is neither reproduced nor claimed here. What *is* claimed, and verified locally, is the attractor structure ([**DERIVE**]): $F(x) = \alpha x^{-5} - x^{-4}$ has the unique positive root $x^* = \alpha$, with

$$F'(x^*) = -\alpha^{-5} = -(\pi/2)^5 \approx -9.5631 < 0,$$

and since $F > 0$ on $(0, x^*)$ and $F < 0$ on (x^*, ∞) , *every* positive initial radius converges to x^* —selection with a basin equal to the whole half-line, not a delicate balance. Numerically, trajectories from $x_0 \in \{0.2, 0.5, 1, 2, 5\}$ all land on $x^* = 0.636620$ to 10^{-15} , and the slope matches $-(\pi/2)^5$ to 10^{-12} (`unjam_inflow.py`, check [3]).

Remark (Why this matters for gate G-S). The extremum (this pillar), the self-averaging of the modulus (`rigid_shell.py`), and this attractor are three independent legs under one statement—*stiffness selects a definite size*—and they are collected as such in the G-S gate of Section 11.

8.3 From Navier–Stokes to the selection normal form

The dispersion (21) is not an ansatz divorced from fluid dynamics. Starting from two-dimensional incompressible Navier–Stokes in vorticity form with a linear drag α and a spectrally rotating body force (an annulus $k \in [k_{\text{low}}, k_{\text{high}}]$ whose excited azimuthal sector rotates in time), one linearizes about rest, projects onto the annulus, and performs a centre-manifold / normal-form reduction of Newell–Whitehead–Swift–Hohenberg type near the band centre k_0 . The amplitude equation that results,

$$\partial_t \phi = (\mu - \sigma_0 k_0^4) \phi + (2\sigma_0 k_0^2) \nabla^2 \phi - \sigma_0 \nabla^4 \phi + \dots,$$

reduces to (21) after rescaling. Physically the viscous and drag channels supply the binding (k^2) slot, while the curvature/geometry of the rotating spectral sector and the injected power shape the penalty (k^4) slot. The selected length is thus inherited by the full nonlinear flow from the linear arrangement of these channels; the nonlinearity saturates the amplitude but does not move the peak.

8.4 The selection is real in the nonlinear pattern

Proposition 4 is linear. The configurational claim is that the *nonlinear* field actually locks onto k_* and inherits the one-half exponent—i.e. that selection survives saturation. We test this directly with a two-dimensional Swift–Hohenberg integration (the minimal nonlinear realization of (21)) and, independently, with the spectrally-rotating DNS pipeline.

Reproducibility box: Pillar III — length selection and the one-half law (`length_selection.py`)

- **Analytic extremum, μ -independence.** For $\mu \in \{-0.5, 0, +0.5\}$ the numerical maximizer of (21) matches $k_* = \sqrt{\varepsilon/2\sigma}$ to 10^{-3} , identically in μ . **PASS**
- **Nonlinear pattern selection.** A saturated 2D Swift–Hohenberg pattern (IF-RK2, several wavelengths per box) locks its spectral peak onto k_* to within 0–7% across $\sigma/\varepsilon \in [0.15, 1.0]$. **PASS**
- **Measured exponent.** Fitting $\log L_*$ versus $\log(\sigma/\varepsilon)$ over the nonlinear runs gives slope ≈ 0.46 , converging to the analytic $1/2$ as binning/resolution improve. **PASS**

Companion DNS (spectrally rotating forcing, ETDRK4, 2/3 dealiasing, $N = 512$): the pre-specified blind pipeline returns $\hat{\beta} \approx 0.501$ with a 95% cluster-bootstrap interval containing $1/2$, stable across forcing sectors, resolutions, and robust slope estimators (OLS, Theil–Sen, errors-in-variables). Run `python length_selection.py` for the first three checks.

The strength of the evidence. Three independent facts agree: the exponent is $1/2$ by a one-line calculus argument, a nonlinear pattern reproduces it, and a forced DNS with a deliberately blind analysis pipeline recovers 0.501. A reasonable null—that the “selected” length is an artifact of the analysis—is excluded by the shuffle test in the DNS pipeline, which returns a slope near zero when the early/late windows are re-paired at random. The selected length is a property of the flow, and its exponent is the exponent of the arrangement’s geometric mean.

Pillar III, in one line. A binding-versus-penalty arrangement selects $L_\star \propto (\sigma/\varepsilon)^{1/2}$ with an exact one-half exponent independent of growth offset ([DERIVE]); the nonlinear pattern and a blind DNS both recover it ($\hat{\beta} \approx 0.501$).

9 Pillar IV — Dissipative: arrangement sets dissipation

Events carry a nonnegative dissipation measure with the exact budget $\langle \varepsilon_{\text{inj}} \rangle = \langle \varepsilon_\nu \rangle + \langle \varepsilon_{\text{events}} \rangle$; as $\nu \rightarrow 0$ dissipation migrates to the event channel and saturates at the injection rate. At substrate level the events are local *unjamming avalanches* of the marginal arrangement; driven marginality (G-SOC) is the gated reason the channel is always available. Their sizes obey a scale-free law with the marginal-stability exponent ($\tau \rightarrow 1.46$), measured directly in developed turbulence. On release into decay the dissipation coefficient C_ε follows the Vassilicos non-equilibrium law $\propto \text{Re}_\lambda^{-1}$ and relaxes onto the fixed-point plateau (the zeroth law).

The last pillar addresses the fact most often presented as a property of a *limit* rather than of the medium: anomalous dissipation. We give it a configurational origin. Rearrangement events carry energy out of the resolved flow as a nonnegative measure, and in a forced steady state energy conservation forces this event dissipation to take over from the viscous channel as viscosity vanishes. The dissipation rate is then set by the large-scale arrangement (the injection), not by the microscopic viscosity.

9.1 Events as a dissipation measure

Event dissipation (locked record, derived budget)

[LOCK] (A0, event part). Each rearrangement event k at $(\mathbf{x}_k, t_k)$ releases a binding energy $\Delta E_{\text{bind},k} \geq 0$. The space–time event measure

$$\mu_{\text{events}} = \sum_k \Delta E_{\text{bind},k} \delta_{(\mathbf{x},t)=(\mathbf{x}_k,t_k)} \geq 0$$

defines a windowed event dissipation rate $\varepsilon_{\text{events}} = \langle d\mu_{\text{events}}/(dt d\mathbf{x}) \rangle_W$. **[DERIVE] (budget).** In a steady window with periodic or controlled boundaries and a fixed injection $\langle \varepsilon_{\text{inj}} \rangle = I$, energy conservation gives the exact budget

$$\langle \varepsilon_{\text{inj}} \rangle = \langle \varepsilon_\nu \rangle + \langle \varepsilon_{\text{events}} \rangle, \quad \langle \varepsilon_\nu \rangle = 2\nu \langle Z \rangle. \quad (24)$$

Equation (24) is the windowed version of the exact identity (16) verified in Pillar I, with the single addition of a nonnegative event channel on the right. It is bookkeeping: whatever energy enters must leave, through one channel or the other.

9.2 What the events are: local unjamming of the marginal substrate

Up to here, “event” has been operational: a rearrangement that releases binding energy, $\varepsilon_{\text{bind}} = \Theta\sigma \geq 0$. The substrate section gives the operational definition a microscopic face, at no new

cost.

[DERIVE] (geometry) — events are local unjamming avalanches

At the margin the shear reserve of the packing is its overlap distribution, and it vanishes with the distance to isostaticity (Section 3; `unjam_inflow.py`). A local shear demand exceeding the local reserve *must* break contacts; broken contacts release stored contact (binding) energy; and because the medium is marginal, one break lowers neighbouring reserves and propagates—an *avalanche* of unjamming that re-jams when the released stress is absorbed. That is an event in precisely this pillar’s sense: localized in space–time, dissipative, with $\varepsilon_{\text{bind}} \geq 0$ guaranteed because contact energy is nonnegative and breaking only releases it. The identification is geometric bookkeeping, not a new postulate.

Why the event channel is *always available* in a driven medium is the substrate gate G-SOC (§3.6): drive unjams, rest re-jams, and the companion observes the resulting pinning at marginal coordination ($g^*/g_0 \approx 1.1\text{--}1.3$). If G-SOC fails—a steady state drifting with drive instead of pinning—this identification loses its “permanently marginal” premise; the falsification route is stated there.

9.3 The anomaly as channel migration

Claim 1 (Onsager-type anomaly via arrangement). [DERIVE] (budget level). Consider a one-parameter family in which the viscous channel is weakened ($\nu \rightarrow 0$) while the event-core scale r_c shrinks accordingly. If the steady budget (24) holds and the event channel can absorb the flux, then

$$\langle \varepsilon_\nu \rangle \rightarrow 0, \quad \langle \varepsilon_{\text{events}} \rangle \rightarrow I > 0,$$

so the total dissipation does *not* vanish with viscosity but saturates at the injection rate I . [GATE] (mechanism). The premise that the event channel *physically carries the inertial flux down to r_c* —that event activity concentrates where velocity increments violate the $1/3$ Hölder threshold, and that this concentration persists under resolution—is exact in 1D Burgers, resolution-stably demonstrated in 2D, and resolution-converged in developed 3D turbulence (Section below and `ns3d.py`); only its infinite-Reynolds asymptotic form remains an open gate.

The split is the content. In the standard field account, anomalous dissipation is the statement that the inviscid limit retains a defect; here the defect is named—it is the event measure—and the budget shows why its magnitude is fixed by the large scale. *The dissipation is configured:* it is what the arrangement must shed to stay steady, independent of how small the viscosity is.

9.4 The metriplectic decomposition, derived

The saturation is not an assumption; it follows from the structure of the evolution once dissipation is admitted alongside the Hamiltonian core of Section 4. Write the rate of change of any observable A as a reversible (Poisson) part plus an irreversible (metric) part,

$$\frac{dA}{dt} = \{A, H\} + [A, S], \quad (25)$$

with the degeneracy conditions $\{A, S\} = 0$ and $[A, H] = 0$ (the metriplectic, or GENERIC, conditions). Here H is the *total* energy: the reversible bracket conserves entropy, the irreversible bracket conserves the total energy, and $G\nabla C = 0$ keeps the Casimirs (enstrophy in 2D, helicity in 3D) constant. The observable that actually dissipates is the *resolved* (kinetic) energy E , which is not H : the metric bracket conserves the total H by *transferring* resolved energy into unresolved internal and binding modes. Its reversible transport—exactly the U_2+U_3 Hamiltonian flow of

(1)–(2)—conserves E , while its irreversible part can only drain it, $dE/dt|_{\text{irr}} \leq 0$, with $dS/dt \geq 0$ by positive semidefiniteness.

Apply this to the resolved energy of a forced steady state, with the two dissipative transfer channels—a viscous one with coefficient $a\nu$ and an event (binding) one with coefficient b , both nonnegative by the semidefiniteness—and an injection I :

$$\frac{dE}{dt} = \underbrace{\{E, H\}}_{=0} + I - a\nu E - bE \implies E_\star = \frac{I}{a\nu + b}. \quad (26)$$

The two dissipation rates are then $\varepsilon_\nu = a\nu E_\star$ and $\varepsilon_{\text{bind}} = bE_\star$, and they close the budget exactly, $\varepsilon_\nu + \varepsilon_{\text{bind}} = I$, recovering (24). The inviscid limit is now immediate:

$$\nu \rightarrow 0: \quad E_\star \rightarrow \frac{I}{b}, \quad \varepsilon_\nu \rightarrow 0, \quad \varepsilon_{\text{bind}} \rightarrow I. \quad (27)$$

The viscous channel switches off and the event channel takes over the *entire* injection—channel migration, not a coincidence of limits. The binding rate is the event law of Section 4, $\varepsilon_{\text{bind}} = \Theta\sigma$ with $\sigma \geq 0$, so the same nonnegativity that makes the metric bracket dissipative makes $\varepsilon_{\text{bind}} \geq 0$. Crucially, a *naive* closure that lets cores merge by contact *without* the metric (binding) channel fails to close the budget—it leaves a residual of order the injection—which is exactly why the event channel is necessary and is the gate the numerics probe.

9.5 What reproduces, and what is deliberately left open

We separate cleanly the part that is a theorem (the budget and its saturation) from the part that is a gate (that a microscopic event rule realizes the flux). An honest negative result sharpens the distinction: a naive position-based merger rule, in which like-sign vortices simply bind on contact, *fails* to close the budget (residual ≈ 0.49). Saturation is therefore not automatic; it requires the event channel to be energy-consistent, which is exactly what the gate demands.

Reproducibility box: Pillar IV — budget closure and saturation (metriplectic_vortex.py, validate_all.py)

Energy-consistent reduced metriplectic vortex-gas model: forced steady state with a smooth viscous sink and a binding-event sink, $\dot{E} = I - a\nu E - bE$, giving $E_\star = I/(a\nu + b)$.

- **Budget closes exactly.** $\langle \varepsilon_{\text{inj}} \rangle = \langle \varepsilon_\nu \rangle + \langle \varepsilon_{\text{bind}} \rangle = I = 0.21$ for every ν . **PASS**
- **Saturation.** As ν falls from 0.05 to 5×10^{-4} , $\langle \varepsilon_\nu \rangle \rightarrow 0$ (from 0.032 to 4×10^{-4}) and $\langle \varepsilon_{\text{bind}} \rangle \rightarrow I$ (from 0.178 to 0.2096). **PASS**
The plateau equals the injection rate I by construction: the dissipation is set by the large-scale arrangement, not by ν .
- **Reference viscous branch.** The viscous curve ($\text{Re}_{\text{eff}}, \max_t \varepsilon_\nu$) that the event branch overlays is the validated output of Pillar I’s solver, with $\max_t \varepsilon_\nu = 2\nu Z(0) \propto \nu$ exactly. **PASS**
- **Negative control.** A naive contact-merger rule does *not* close the budget (residual ≈ 0.49): saturation is conditional, matching the **[GATE]** status of the mechanism. (documented, not a PASS)

Run `python validate_all.py`; checks 4–5 are the budget and saturation.

The mechanism gate—that the event channel *physically carries* the inertial flux to the dissipation scale—is the heaviest open claim. In one dimension it is exactly tractable, and we test it directly: in Burgers turbulence the events are shocks, and the Duchon–Robert defect is exact.

Reproducibility box: Pillar IV mechanism — events carry the flux (1D Burgers, `event_flux.py`)

Decaying Burgers from $u = \sin x$; once the shock forms it is the rearrangement event. The inviscid energy defect of a shock of jump Δu is $(\Delta u)^3/12$, localized at the shock.

- **Dissipation = event flux.** The viscous dissipation $\int \varepsilon_\nu$ equals $(\Delta u)^3/12$ to $\sim 10\%$ (the residual is the finite-window jump estimate), across ν . **PASS**
- **ν -independence.** $\int \varepsilon_\nu = 0.61, 0.64, 0.65, 0.66$ as ν falls $0.02 \rightarrow 0.0025$ (a factor 8): the dissipation is fixed by the event, not by viscosity. **PASS**
- **Flux concentrates at the event.** The Duchon–Robert inter-scale flux $\Pi_\ell(x) = \frac{1}{4} \int \phi'_\ell(r) \delta u(r)^3 dr$ is $\sim 100\%$ localized at the shock, and matches the dissipation at the inertial scale. **PASS**

$\Rightarrow$ in the canonical (1D) case the event *does* carry the inertial flux to the dissipation scale. The same Duchon–Robert structure governs 2D/3D Navier–Stokes (vortex sheets and stretching events); the next box tests it directly in two dimensions.

Reproducibility box: Pillar IV (flux-carrying), 2D Navier–Stokes (`multid_flux.py`)

A forced-scale vorticity field is evolved to a filamented snapshot. In 2D the forward cascade is the *enstrophy* cascade: enstrophy is carried to the small (dissipation) scale, where the energy dissipation $\varepsilon_\nu = 2\nu Z$ lives, and the events are the thin intense vorticity-gradient structures. Filtering at scale ℓ (Germano), the local inter-scale enstrophy flux $\Pi(x) = -\sigma_j \partial_j \bar{\omega}$, with subfilter vorticity flux $\sigma_j = \overline{u_j \bar{\omega}} - \bar{u}_j \bar{\omega}$, is compared against the palinstrophy density $|\nabla \bar{\omega}|^2$ (proportional to the local dissipation).

- **Events carry the flux.** The top 20% most intense gradient regions carry $\approx 60\%$ of the forward enstrophy flux ($\langle \Pi \rangle > 0$); only $\approx 12\text{--}16\%$ of the area carries half of it. **PASS**
- **Flux co-locates with dissipation.** $\text{corr}(\Pi, |\nabla \bar{\omega}|^2) \approx 0.28$: the inter-scale flux concentrates where the dissipation is. **PASS**
- **Resolution-stable.** Across $N = 192, 256, 320$ at fixed physical ℓ , the top-20% fraction holds at 63%, 61%, 58% and the half-flux area at 12%, 14%, 16%. **PASS**

$\Rightarrow$ in 2D the events *do* carry the inter-scale flux to the small scale—as in 1D, but shown by spatial concentration rather than an exact identity. This narrows the gate from “multidimensional” to the *high-Reynolds asymptotic* question—whether the concentration sharpens onto a measure-zero set as $\text{Re} \rightarrow \infty$ —which is beyond reach at these resolutions and is stated honestly as the residual **[GATE]**.

Reproducibility box: Pillar IV (flux-carrying), 3D Navier–Stokes (`ns3d.py`)

The genuine Onsager case: in 3D the forward cascade is the *energy* cascade. A 3D pseudo-spectral solver (velocity/rotational form, 2/3 dealiasing, IF-RK4) is first validated—in the inviscid limit it conserves energy to machine precision ($\Delta E/E \sim 10^{-16}$, the rotational nonlinearity being energy-orthogonal to $\mathbf{u}$) and stays divergence-free to $\sim 10^{-14}$. Low- k forcing then drives a developed cascade, confirmed by the velocity-derivative skewness reaching the canonical ≈ -0.5 . The filtered (Germano) local energy flux $\Pi = -\tau_{ij} \bar{S}_{ij}$ is compared with the strain magnitude $|\bar{S}|^2 \propto$ the local dissipation.

- **Flux is forward.** $\langle \Pi \rangle > 0$, about $0.4\text{--}0.6\varepsilon$ in the inertial range and rising toward ε near the dissipation scale; 94–95% of $|\Pi|$ is forward. **PASS**
- **Events carry it.** At a fixed physical filter scale the top 20% of the strain field carries $\approx 60\text{--}66\%$ of the forward flux; only $\approx 10\text{--}14\%$ of the volume carries half. The flux is *strain*-aligned: $\text{corr}(\Pi, |\bar{S}|^2) \approx 0.80$ (and only weakly vorticity-aligned), exactly as 3D theory predicts. **PASS**
- **Resolution-converged.** Across $N = 64, 80, 96, 128$ (identical physics, fixed physical fil-

ter): top-20% = 63%, 66%, 60%, 60%; half-flux volume = 13%, 10%, 14%, 14%; correlation = 0.80, 0.80, 0.82, 0.80. **PASS**

$\Rightarrow$ in 3D, as in 1D and 2D, the events carry the inter-scale energy flux to the dissipation scale, and the concentration is resolution-converged. The only residual is the *infinite-Reynolds asymptotic*—whether the support tends to measure zero as $\text{Re} \rightarrow \infty$ —which is the genuine open Onsager/Duchon–Robert question, beyond any finite simulation and stated honestly as the surviving **[GATE]**.

A second, independent test of the same identification sharpens it from geometry to a number. If the events are genuinely unjamming avalanches of a *marginal* substrate, their sizes cannot carry an intrinsic scale: the thresholded dissipation structures must be distributed as a power law, with the exponent fixed by the marginal-stability avalanche relation. Measuring the local dissipation field $\varepsilon(\mathbf{x}) = 2\nu S_{ij}S_{ij}$ of developed 3D turbulence, thresholding it, and recording the size distribution of the connected high-dissipation structures is therefore a direct, falsifiable test — and an exponential (any intrinsic scale) would refute the avalanche reading outright.

Reproducibility box: Pillar IV (events are marginal-stability avalanches) — the dissipation-event size law (dissipation_avalanche.py)

From the same validated solver, developed turbulence (skewness ≈ -0.5) is sampled over decorrelated snapshots; the full-resolution dissipation field is thresholded at $h(\varepsilon)$ and the connected structures are labelled with *periodic* boundaries; their volume distribution $P(s)$ is fitted by maximum likelihood (Clauset, discrete, KS-selected cutoff) and cross-checked by logarithmic binning. The marginal-stability (Lin–Wyart) relation $\tau = 2 - \frac{\theta}{1+\theta} \frac{d}{d_f}$, with stationary $\theta \approx 0.5\text{--}0.6$ and sheet-like $d_f \approx 2$, predicts *in advance* $\tau \approx 1.4\text{--}1.5$.

- **Scale-free, not exponential.** $P(s)$ is a power law over more than a decade (log-binned $R^2 \approx 0.97\text{--}0.99$), and a fitted exponential is rejected at every threshold (positive power-law/exponential log-likelihood ratio). **PASS**
- **Sheet-like structures.** The gyration-radius–volume scaling gives $d_f \approx 2.0\text{--}2.2$, resolution-stable — the geometric input the prediction assumes. **PASS**
- **Predicted exponent, with the right trend.** The measured exponent falls with resolution into the predicted band: $\tau = 1.60 \pm 0.01$ at $N = 64 \Rightarrow \tau = 1.46 \pm 0.01$ at $N = 96$ (log-binned ≈ 1.4), landing inside $[1.4, 1.5]$ along a clean finite-Reynolds trend. **PASS**

$\Rightarrow$ the dissipation events of developed turbulence are distributed as marginal-stability avalanches *with the predicted exponent*, tying Pillar IV’s events to the substrate of Section 3 by a quantitative law and not only by a shared name. This is a genuine prediction (not a postdiction): the band $[1.4, 1.5]$ was fixed by the substrate before the turbulence was measured. The residual is the one carried throughout—the precise $\text{Re} \rightarrow \infty$ exponent, beyond any finite resolution, remains a **[GATE]**.

A third developed-turbulence test closes the dissipation story by its *magnitude*. The budget box showed the dissipation saturates at the injection rate (the anomaly); the event-RG (§5) reads that saturated, scale-independent state as the Π_L fixed point. The sharp consequence concerns *leaving* that state: out of equilibrium the dissipation coefficient $C_\varepsilon = \varepsilon L/u'^3$ must not stay constant but flow as the non-plateau RG transient. Releasing a forced field into free decay (the Goto–Vassilicos protocol) and tracking C_ε against the Taylor-scale Reynolds number Re_λ tests exactly this — and a constant C_ε throughout would refute the “non-plateau transient” reading.

Reproducibility box: Pillar IV (dissipation magnitude) — non-equilibrium C_ε and the fixed-point plateau (nonequilibrium_dissipation.py)

A developed field (skewness ≈ -0.5) is released into free decay; the integral scale L (from the energy spectrum), u' , $\varepsilon = 2\nu Z$, and $\text{Re}_\lambda = u'\lambda/\nu$ are logged along the decay, giving the trajectory $C_\varepsilon(\text{Re}_\lambda)$. Equilibrium (Taylor’s zeroth law = the Π_L fixed point) predicts $C_\varepsilon = \text{const}$ (slope 0); Vassilicos’s non-equilibrium law predicts $C_\varepsilon \propto \text{Re}_\lambda^{-1}$.

- **Non-equilibrium law, not constant.** In the high- Re_λ (early-decay) window C_ε rises as Re_λ falls, with slope $C_\varepsilon \propto \text{Re}_\lambda^{-1.01 \pm 0.02}$ at the well-resolved $N = 96$ ($k_{\max}\eta > 1.3$ throughout, $R^2 = 0.98$) and -1.14 at $N = 64$ — the Vassilicos -1 , *opposite* to the equilibrium constant. **PASS**
- **Relaxation to the fixed point.** As the decay becomes self-similar (low Re_λ) the slope flattens to ≈ -0.3 and C_ε approaches a plateau (≈ 1.0): the non-plateau transient relaxes onto the Π_L fixed point, i.e. the zeroth law itself. **PASS**

$\Rightarrow$ the framework’s reading of anomalous dissipation is borne out dynamically: the constant- C_ε state *is* the event-RG fixed point (§5), and *departure* from it follows the predicted non-plateau flow with the Vassilicos exponent. The residual is the universality of the precise exponent and plateau value at higher Re , beyond these resolutions — the same infinite-Reynolds **[GATE]**.

Pillar IV, in one line. Rearrangement events carry a nonnegative dissipation measure whose steady-state budget is exact (**[DERIVE]**) and saturates at the injection rate as $\nu \rightarrow 0$ (Onsager-type anomaly); that the event channel *physically* carries the inertial flux to the dissipation scale is *exact in 1D Burgers, resolution-stably concentrated on events in 2D*, and *resolution-converged on strain events in developed 3D turbulence*, leaving only the infinite-Reynolds asymptotic limit as a **[GATE]**. Independently, those events are distributed as *marginal-stability avalanches*: the thresholded dissipation structures follow a scale-free size law with the pre-registered Lin–Wyart exponent ($\tau \rightarrow 1.46 \in [1.4, 1.5]$), not an exponential. And by *magnitude*: on release into decay the dissipation coefficient flows as the Vassilicos non-equilibrium law $C_\varepsilon \propto \text{Re}_\lambda^{-1}$ (-1.01 at the resolved $N = 96$) and then relaxes onto the constant- C_ε plateau — the Π_L fixed point, i.e. the zeroth law.

10 Reach and universality: the same mechanism in foreign domains

The same mechanisms run unchanged in foreign domains: Burgers dissipation saturates as $\nu \rightarrow 0$ (the events are shocks), the capacity identity holds in $d = 2\text{--}5$ to 10^{-16} , and a Turing reaction–diffusion system locks onto the dispersion-selected $k_\star$ to 6–8%.

Before deepening any single pillar, we ask the prior question: how far does “form follows arrangement” reach? A thesis that explains only fluids is a curiosity; a thesis whose mechanisms recur, unchanged, in unrelated domains is a foundation. We therefore *transplant* each of the three substantive mechanisms into a domain it was not built for, and ask whether it survives. It does. Crucially, the underlying mathematical structures are ones physics already knows to be universal; the contribution of the configurational reading is to show that they are *the same statement*.

10.1 Dissipation transplanted: Burgers shocks are events

The cleanest test of Pillar IV outside vortex dynamics is one-dimensional Burgers turbulence, where anomalous dissipation is classically exact. A shock is a rearrangement event: the field reorganizes discontinuously and sheds energy. The configurational prediction is that the dissipation rate is set by the shock, not by the viscosity, so it must *saturate* as $\nu \rightarrow 0$.

Reproducibility box: Universality U1 — Burgers anomalous dissipation (universality.py)

Decaying Burgers $u_t + uu_x = \nu u_{xx}$ from $u = \sin x$ (a single shock forms at $t = 1$); viscous dissipation $\varepsilon_\nu = \nu \langle u_x^2 \rangle$ measured at $t = 1.6$. As ν falls $0.04 \rightarrow 0.006$ (a factor ~ 7): $\varepsilon_\nu = 0.088, 0.098, 0.102, 0.104$. **PASS**

The dissipation does *not* fall with ν ; it saturates toward the inviscid shock value (relative spread 6% over the small- ν runs). The dissipation is set by the event (the shock), exactly as Pillar IV claims.

This is not a coincidence of the model. The defect-measure formulation of anomalous dissipation—a nonnegative measure that appears in the energy balance of rough Euler/Navier–Stokes limits, concentrating at singular events—is precisely the object Pillar IV calls the event measure μ_{events} . Burgers shows it where it is exactly computable; the same structure (defect concentrated on shocks, vortex sheets, reconnection) is what the field literature documents in compressible and three-dimensional turbulence. The VP reading gives that defect a configurational origin: it is the energy released when the arrangement rearranges.

10.2 Geometry transplanted: the capacity identity is dimension-general

Pillar II was stated for two-dimensional cores. But the identity is measure-accounting, and accounting does not know the dimension. In d dimensions, with V_d the volume of the unit ball,

$$N_\star r_{\text{eff}}^d = \phi s L_{\text{core}}^d, \quad r_{\text{eff}}^d = \frac{1}{N_\star} \frac{1}{V_d} \sum_k \text{vol}_k. \quad (28)$$

Reproducibility box: Universality U2 — capacity identity in dimension d (universality.py)

Synthetic packings, 40 blobs, core shape factor $s = 0.6$. Identity residual of (28): $d=2: 1.4 \times 10^{-16}$; $d=3: 0$; $d=4: 1.6 \times 10^{-16}$; $d=5: 0$. **PASS**

Replacing r_{eff} by a single-core measurement and varying the packing from single-scale to heavy-tailed reproduces the fluid audit’s signature: median residual grows from $\sim 150\%$ to $\sim 400\%$ and the mean-minus-median gap widens (multiscale arrangements are large and skewed).

Equation (28) is the integer-packing case of the Minkowski–Bouligand (box-counting) relation $N(r) \sim (L/r)^d$ that defines fractal dimension. The RCCI residual is therefore a coarse multifractality probe: small for arrangements that fill a single scale, large and heavy-tailed for those that do not. “Arrangement fixes length” is geometric universality—true in any dimension, with the failure of one length being the signature of fractal, multiscale packing.

10.3 Selection transplanted: chemistry selects the same length

Pillar III’s binding-versus-penalty selection was demonstrated in a Swift–Hohenberg flow surrogate. The strongest universality statement is that the k^2 – k^4 dispersion is the *generic normal form* near any finite-wavelength instability, so a system of entirely different physical origin—chemistry—must select its length the same way.

Reproducibility box: Universality U3 — reaction–diffusion Turing selection (universality.py)

Schnakenberg reaction–diffusion (a chemical morphogenesis model), $a = 0.1$, $b = 0.9$, $\gamma = 80$. The spontaneously selected wavelength of the Turing pattern matches the linear-dispersion maximizer $k_\star$: $d=35: k_\star^{\text{lin}} = 3.87$ vs $k_\star^{\text{pat}} = 3.56$ (8%); $d=45: 3.67$ vs 3.44 (6%). **PASS**

A chemical system, with no fluid in it, selects the dispersion-predicted length.

The Swift–Hohenberg normal form generically captures stationary finite-wavelength pattern formation: convection rolls, Turing patterns in morphogenesis, Faraday waves, buckling sheets, directional solidification, nonlinear optics, and even segregation in flowing granular media all reduce to it near onset, and in each the selected band scales as the *square root* of the control parameter—the one-half law of Proposition 4. “Arrangement selects length” is as broad as pattern formation itself; the configurational reading is that the selected length is always the geometric mean of the arrangement’s binding and penalty scales.

10.4 What the reach test establishes

Universality finding

The three substantive mechanisms are not three borrowings from three fields. Each is an instance of a structure physics already treats as universal—the pattern-selection normal form (dynamical), box-counting/measure-accounting (geometric), and the anomalous-dissipation defect measure (dissipative)—and the configurational postulate A0 is what makes them *one statement seen three ways*. The reach of “form follows arrangement” is therefore the union of pattern formation, packing geometry, and anomalous dissipation; its content is the claim that these are the same phenomenon, because each is what an arrangement must do.

Honest limits of reach. (i) The framing reinterprets; it does not replace the operational tools (Swift–Hohenberg, box-counting, Duchon–Robert remain the calculators). (ii) Selection requires a genuine finite-wavelength instability—a binding/penalty competition; systems with monotone dispersions or pure cascades do not select a length this way. (iii) The dissipative universality is established at the level of the *defect measure* (a phenomenological certainty), while the configurational *mechanism*—that rearrangement events carry the flux to the dissipation scale—remains the open [GATE] of Section 14. The reach is wide; the frontier is the mechanism.

11 Cross-scale extensibility: one arrangement law from genomes to vortices

One arrangement law spans scales: $Nv_{\text{eff}} = \phi V_{\text{core}}$ from genomes to vortices, droplet selection at $k_*a = 0.697$, a plane breathes while a sphere closes—and Mechanism M now starts at Step 0, rotation unjams the margin. The quantum bridges G-Q and G-S remain graded gates with carried-out falsification legs.

Section 10 showed the mechanisms reach across *domains* at a fixed scale. This section asks whether they reach across *scales*—from biomolecular condensates to vortex cores, and conjecturally to the quantum core. The strongest evidence is not an analogy we draw but a convergence we find: an independent line of the author’s work on genomes arrived at the *same* capacity identity, the *same* two control knobs, and the *same* selection principle. We keep the discipline of Principle 1: the demonstrated unifications are graded [DERIVE], the cross-scale bridges are graded [GATE], and framework-internal integer coincidences are flagged as such, not asserted as physics.

11.1 The same capacity identity in genomes and in vortices

A two-layer interpretation of DNA models a chromatin region as a phase-separated condensate with a firm interface—a “stiffness shell.” Its packing obeys, in that work’s own notation,

$$N v_{\text{eff}} \approx \phi V_{\text{core}}, \quad (29)$$

with N units of effective volume v_{eff} at fill fraction ϕ in a core of volume V_{core} . Writing $v_{\text{eff}} = V_d r_{\text{eff}}^d$ and $V_{\text{core}} = s V_d L_{\text{core}}^d$, (29) becomes $N r_{\text{eff}}^d = \phi s L_{\text{core}}^d$ —*exactly* the rotational core capacity identity of Pillar II (Eq. (18)), in $d = 3$. This is not a resemblance; it is the same equation, reached independently from sequence data.

Reproducibility box: Extensibility E1 — the DNA condensate identity is the fluid RCCI (extensibility.py)

Synthetic $d=3$ packing: (29) and the RCCI form $N r_{\text{eff}}^3 = \phi s L_{\text{core}}^3$ are both satisfied to residual 0 (same equation). The condensate “preferred size” is the minimum-residual-stress fill: modeling over-fill as steric clash and under-fill as interfacial/void cost, the minimum sits at $\phi^* \approx 0.64$ —the random-close-packing (jamming) point. **PASS**

$\Rightarrow$ one arrangement law governs biomolecular condensates and vortex cores; “a droplet has a preferred size” and “a vortex core has a fixed radius” are the same statement.

The condensate’s two knobs are the interfacial tension γ (the firmness of the shell, set by composition) and the fill fraction ϕ (set by how much material is packed). These are exactly the ϕ and shape factor s of the RCCI, with γ supplying the “firmness” that the fluid audit reads as the cost of a misshapen interface. *Jamming and fluid are then two phases of one arrangement*: the preferred fill sits at the jamming point $\phi_c \approx 0.64$, below which the medium flows and above which it clashes. This is the precise sense in which jamming theory and fluid dynamics are not different subjects. The jamming point invoked here is, moreover, not folklore: the companion bundle *measures* it for the substrate ($\phi_{\text{jam}} = 0.633$ at the isostatic onset $z \rightarrow 2d$, with no physical constant in the input; concept DOI 10.5281/zenodo.17932566). Inheriting the companion’s own discipline, we flag explicitly that the numeral adjacency of $\phi_{\text{jam}} \approx 0.633$ and $2/\pi = 0.6366$ is *not* claimed as an identity, a cause, or evidence anywhere in this document (Section 14).

11.2 Droplet formation is length selection

The condensate picture also closes the loop with Pillar III. A droplet does not merely have a preferred size statically; a liquid jet *selects* a droplet size dynamically, by the Rayleigh–Plateau instability, whose dispersion is set by the interfacial tension γ .

Reproducibility box: Extensibility E2 — droplet size is selected by a dispersion maximum (extensibility.py)

Rayleigh’s inviscid dispersion for a liquid cylinder of radius a , $\omega^2 \propto x(1-x^2)I_1(x)/I_0(x)$ with $x = ka$, peaks at $x_* = 0.6970$ (classical 0.697), giving droplet spacing $\lambda_* = 9.014a$ (classical $9.01a$). **PASS**

$\Rightarrow$ “how a droplet forms” is the binding-versus-penalty length selection of Pillar III, with the interfacial tension γ playing the binding role. Selection, capacity, and condensate interface are one mechanism.

11.3 Why a planar vortex breathes but a spherical core closes

The author’s framework distinguishes the macroscopic planar vortex (a typhoon, which takes air in and pushes it out) from a spherically symmetric core (which closes on itself). This distinction is not interpretive; it follows from incompressible continuity alone.

Reproducibility box: Extensibility E3 — through-flow geometry, 2D vs 3D
(extensibility.py)

Planar (cylindrical, axisymmetric): $\frac{1}{r}\partial_r(ru_r) + \partial_z w = 0$. A radial inflow $u_r = -Ar$ forces an *axial* updraft $w = 2Az$ —the eye updraft / Ekman pumping. The perpendicular z -axis is the exhaust route, so the vortex sustains a steady intake-and-exhaust. *Spherical* (full symmetry): $\frac{1}{R^2}\partial_R(R^2u_r) = 0 \Rightarrow u_r \propto R^{-2}$, a pure source/sink with no perpendicular axis; a steady inflow has nowhere to go and must close. **PASS**
 $\Rightarrow$ the same rotating-inflow arrangement is a sustained through-flow in two dimensions and self-closing in three—exactly the stated typhoon/quantum-core contrast.

11.4 The co-rotation mechanism: how small rotations build a large one

Before the conjectural bridges, we isolate the *mechanism* that the C_3 lattice structure actually encodes—and that the bridges rely on; we refer to this chain as *Mechanism M*. It is not a claim about how many vortices a typhoon contains; it is the chain by which small rotations are *forced* to co-rotate, and how that forced co-rotation builds a large-scale rotation with inflow and outflow. Each step is demonstrable.

Step 0: the margin makes rotation an unjamming switch. Before frustration can do anything, rotation must be able to move material at all. Section 3 supplies exactly this, and it is inherited geometry, not a postulate: a *rigid* rotation costs a packing nothing (every bond extension vanishes identically, $\hat{\mathbf{n}} \cdot (\hat{\mathbf{z}} \times \mathbf{r}) = 0$), while any *differential* rotation is a shear—and at the margin the shear reserve is the vanishing overlap distribution, so the rotation breaks contacts and unjams. Rotation is therefore the *fluidization switch* of the substrate, and the directed inflow the steps below pump has a geometric origin: the unjammed material moves, and the one open direction is inward along the rotation’s own demand.

Reproducibility box: Mechanism M, Step 0 — rotation unjams the margin
(unjam_inflow.py)

- **Rigid rotation is free:** maximal bond extension per unit rotation under $\mathbf{u} = \theta \hat{\mathbf{z}} \times \mathbf{r}$ is 2.8×10^{-17} —zero to machine precision. **PASS**
- **The reserve dies at the margin:** the contact-breaking shear strain falls with pressure across the ensemble—quantile q_{05} ratio $5.9\times$ between the densest and the most marginal packings, $\text{corr}(\log q_{05}, \log P) = 0.96$, $\text{corr}(\log \text{med}, \log P) = 0.99$. (The very first break is an extreme-value statistic and is reported per packing for honesty; the physical reserve is the overlap *distribution*, and real unjamming is an avalanche of breaks, not a single bond.) **PASS**

At $P \rightarrow 0$ any rotational shear exceeds the reserve: rotation unjams.

Step 1: frustration forces co-rotation. Meshing rotors prefer to *counter*-rotate, like gears. A globally consistent counter-rotation is exactly a proper two-colouring of the contact graph, which exists if and only if the graph is bipartite—no odd cycles. The triangle, the minimal C_3 , is an odd cycle and is *not* two-colourable: counter-rotation is frustrated, so the rotors are forced to *co-rotate*, and one contact is left unsatisfiable. That unsatisfiable residual is the “+1” nozzle of the $82 = 81 + 1$ core. This is the mechanical content of “a circle cannot be cleanly three-divided.”

Step 2: co-rotation builds a larger rotation. Once small rotations are forced to share a sign, their circulations *add* ($\sum_k \gamma_k$) rather than cancel, and like-sign vortices *merge*: two co-rotating cores fuse into a single larger core, whereas opposite-sign cores do not. The forced co-rotation therefore grows a large-scale rotation out of many small ones.

Step 3: the large rotation forces a through-flow. A coherent rotation Ω over a boundary cannot stay purely azimuthal: in the friction (Ekman) layer the rotation is reduced, the radial pressure balance is broken, and a radial *inflow* develops; by continuity (Section 11, E3) that inflow turns into an axial *outflow*. The inflow transport scales with the Ekman depth $\sqrt{\nu/\Omega}$. This is the eye updraft of a cyclone and the tea-leaf paradox: rotation pumps.

Reproducibility box: Mechanism M — forced co-rotation $\rightarrow$ large rotation $\rightarrow$ through-flow (`corotation.py`)

- **Frustration (graph two-colouring).** Square lattice: 0 frustrated edges (bipartite) $\Rightarrow$ counter-rotation, net circulation 0. Triangular lattice: 48 frustrated edges (not bipartite) $\Rightarrow$ co-rotation forced, net circulation $\neq 0$. The C_3 triangle has exactly one frustrated contact (= the +1 nozzle). PASS
- **Merger (2D Navier–Stokes).** Two strongly overlapping equal vortices: co-rotating $\rightarrow$ one connected core (merged); opposite-sign $\rightarrow$ two cores. PASS
- **Ekman pumping.** Radial inflow transport $\propto \Omega^{-1/2}$ (measured slope -0.500) and $\propto \nu^{+1/2}$ (measured $+0.500$): rotation forces an axial through-flow. PASS

The chain is scale-free: the same forced-co-rotation-with-nozzle operates on the quantum lattice (the 82 core) and in macroscopic rotating flows.

The remaining connections the framework proposes are genuine conjectures. We state them as such, with falsification routes, and we do not present their numerical coincidences as established.

[GATE] G-Q: the quantum core as a closed planar vortex

The companion physics whitepaper *derives* (does not posit) a lattice core of count $82 = 81 + 1$: a circle cannot be cleanly three-divided, so the only lattice symmetry compatible with a rotational quantum is the C_3 about a body diagonal, and the three-division packing $3^4 = 81$ (equivalently the lattice-shell count $\#\{R^2 \leq 6\} = 81$) is forced, with the next shell $R^2 = 7$ *empty* by Legendre’s three-square theorem. The remaining “+1” is the single irreducible central residual that does not three-divide—read physically as the *inflow nozzle* (the same “1” as in the shell $7 = 1 + 6$). The conjecture is *not* that a typhoon contains three vortices; it is that the same *mechanism* of Section 11.4 operates here—small rotations frustrated into forced co-rotation, building a large rotation with a nozzle that drives through-flow. **Status.** The lattice combinatorics ($82 = 81 + 1$, $R^2 = 7$ empty) are a genuine theorem (Legendre), forced within the framework’s C_3 rectification lock—not a numerical coincidence; the rotational-unjamming origin of the nozzle’s directed inflow is inherited as derived geometry (Section 3, Step 0); the co-rotation mechanism itself (Steps 0–3) is *demonstrated* (Mechanism M); and the dimensional half (a sphere closes, a plane breathes) is *derived* (E3). What remains [GATE] is the cross-scale *identification* of the quantum lattice’s +1 nozzle dynamics with the macroscopic pumped vortex. **Falsification route.** Build the lattice-with-inflow model and test whether the +1 nozzle term closes in $d = 3$ and sustains a through-flow in $d = 2$ as the continuity geometry (E3) requires.

We carried out that falsification test explicitly, and the mechanism survived it.

Reproducibility box: G-Q falsification test — the 81+1 lattice-with-inflow (lattice_inflow.py)

Build the proton-core lattice: $\#\{R^2 \leq 6\} = 81 = 3^4$ co-rotating cells plus one nozzle (= 82).

- **Forced co-rotation adds.** Biot–Savart over the planar slice (21 cells): co-rotating net circulation = 21 (adds to one large rotation); the counter-rotating control nearly cancels. **PASS**
- **$d = 3$ (closed sphere):** the net radial flux of any regular divergence-free flow through an enclosing sphere is 0 (Gauss + regularity, no external source), so the +1 nozzle *closes*—no steady through-flow. **PASS**
- **$d = 2$ (open axis):** the rotation axis is a through-path; the Ekman radial inflow feeds a finite axial through-flux $Q_z > 0$ —the nozzle is *sustained*. **PASS**

The +1 nozzle sustains a through-flow in two dimensions and closes in three, exactly as E3 requires. Had it done either the reverse, G-Q would be falsified; it did not. This upgrades the *mechanism* (forced co-rotation $\rightarrow$ large rotation $\rightarrow$ dimension-dependent through-flow) to demonstrated; only the *quantum-dynamical identification* remains **[GATE]**.

The self-limited grinder (inherited). The companion bundle also runs the mechanism *forward* as molecular dynamics: starting from 3000 spheres, counter-rotation plus contraction plus incompressibility self-limits to a core of ~ 79 –82 cells whose size is set by the rotation itself, $R_{\text{rms}} \propto 1/\Omega$ —rotation creating the length by unjamming the substrate, i.e. Step 0 run to steady state (concept DOI 10.5281/zenodo.17932566). The companion’s own honesty note is inherited at full strength and must not be under-read: *the MD core is amorphous* ($z_{\text{core}} \approx 8.9$), *so the simulation confirms the mechanism—self-limitation to $\mathcal{O}(82)$ cells with $R \propto 1/\Omega$ —not the lattice type; the exact $82 = 81+1$ at $z = 6$ is the crystalline C_3 count.* The lattice-type identification therefore remains inside the **[GATE]** above, exactly where it was.

[GATE] G-S: rigid-shell birth and the selected quantum length

The companion physics whitepaper’s Section “stiffness selects size; stiffness forces the radius” treats the rotation length $D = \ell_{\text{rot}}$ as a *selected* length, and triangulates it three independent ways,

$$D = 2\lambda_{C,e} = 6\pi^6 r_p = \frac{2\pi\lambda}{A} \approx 4.854 \text{ pm},$$

the third route being a jamming/stiffness selection. The selection law of Pillar III, $L_\star = 2\pi\sqrt{2\sigma/\varepsilon}$, reads σ as a rigidity ($\sim c^2$, the full-jamming elastic speed $c^2 = K/\rho_{\text{eff}}$) and ε as an elasticity (~ 1); the conjecture is that the rigid-shell length so selected is exactly this D , and that the same selection underlies the companion JFM length-selection line. **Status (quantified; three independent legs).** The anchors stand: the two independent empirical/geometric routes $2\lambda_{C,e} = 4.8526 \text{ pm}$ (electron Compton) and $6\pi^6 r_p = 4.8523 \text{ pm}$ (proton radius) agree to $\sim 6 \times 10^{-3}\%$; the jamming-selected route lands at 4.8542 pm, so all three agree to 0.039%—a stiffness-selected length pinned this tightly is the quantitative content of “rigid-shell birth”; and the n -fold law (also forced) gives $\nu_3 = 3\pi^4$ and $m_p/m_e = 6\pi^5$ (−19 ppm). Numbers cited, not re-derived. What this document adds is that the *dimensionless* statement “stiffness selects a definite size” now rests on three independent legs: (i) *self-averaging*—the relative spread of $L_\star \propto \sqrt{\sigma/\varepsilon}$ narrows as the network stiffens (**rigid_shell.py**, detailed below); (ii) *attractor*—the stiffness-versus-inflow balance has a unique, globally stable fixed point with $F'(x^\star) = -(\pi/2)^5$ (Section 8; **unjam_inflow.py**); (iii) *rotation sets the size*—the companion’s molecular-dynamics grinder self-limits with $R_{\text{rms}} \propto 1/\Omega$ (inherited; “the self-limited grinder,” Section 11). What remains **[GATE]** is whether the fluid selection law and the

jamming selection are the *literally same* mechanism, and the specific calibration of A . **Leg (i) in detail (carried out, rigid_shell.py).** The dimensionless content is confirmed: in random spring networks the modulus self-averages as the network stiffens, so the relative spread of the selected length $L_\star \propto \sqrt{\sigma/\varepsilon}$ narrows ($3.4\% \rightarrow 1.8\%$ as the coordination rises $z=4.5 \rightarrow 15$)—the quantitative meaning of rigid-shell birth, a length that sharpens as rigidity grows. **The honest boundary.** The selection law and the JFM exponent are *dimensionless*: they fix the scaling ($1/2$) and the sharpening, not the absolute scale. Reaching the absolute 4.854 pm requires the dimensional anchor $D = 2\pi\lambda/A$, the conceptually difficult dimensionless-to-dimensional mapping of the physics whitepaper; it is referenced, not re-derived here. So this gate is now split: the dimensionless selection and sharpening are **[DERIVE]**; the dimensional anchoring is the residual **[GATE]**.

11.5 The extensibility map

Scale	Arrangement object	What fixes form	Status
Quantum core	82-lattice, rotation+inflow	sphere closes / annihilates	[GATE] + E3
Particle mass	rigid-shell birth ($c^2/1$)	stiff-limit selection	[GATE] + Pillar III
Vortex / typhoon	core arrangement ($L, \phi, s, N_\star$)	RCCI; plane breathes	[DERIVE] (II, E3)
Turbulent dissipation	rearrangement events	defect measure / Onsager	[DERIVE] (IV, U1)
Droplet / condensate	interfacial tension γ , fill ϕ	min residual stress; Plateau	[DERIVE] (E1, E2)
Jamming / soft matter	fill $\phi \rightarrow \phi_c$	rigidity onset (fluid $\leftrightarrow$ solid)	[DERIVE] (E1)
Genome / DNA	stiffness shell = condensate	same identity $Nv = \phi V$	[DERIVE] (E1)

Table 4: Extensibility of “form follows arrangement.” The three mechanisms (capacity identity, interfacial-tension/fill selection, event dissipation) are scale- and substrate-free; the DNA line’s independent convergence on the same capacity identity is the evidence. Rows marked **[DERIVE]** are demonstrated; rows marked **[GATE]** are conjectural bridges to the quantum scale with stated falsification routes.

What the extensibility test establishes. The reach of the thesis is not rhetorical. The capacity identity, the interfacial-tension/fill selection, and the event-dissipation channel recur, with the *same* equations, in condensates, droplets, jammed packings, vortex cores, and genomes—the last reached independently, which is the strongest evidence that the structure is real rather than imposed. The bridges to the quantum core remain conjectures with explicit falsification routes. The honest summary: *the demonstrated reach is from genomes to vortices through one arrangement law; the frontier is whether the same law fixes the quantum length.*

12 The thesis at work: natural form as forced arrangement

Six long-studied forms read as selected arrangements: the GRS size, Saturn’s hexagon, the typhoon as Mechanism M, convective plumes as event flux, jet staircases as a selected spacing $\Delta y \sim \min(L_D, L_\beta)$, and the granular avalanche as flow that is sustained unjamming.

The strongest test of a reframing is whether it reads the visible world without strain. We take four long-studied phenomena and show that each is, in the language of this document, a *selected arrangement*—a form that follows from the forced rotation and the event-RG fixed point of Sections 4 and 5. These are interpretive applications, graded accordingly: the empirical phenomena are untouched, and we claim only that the thesis organizes them coherently, not that it derives them from first principles. The mechanisms invoked are exactly the ones already demonstrated (Mechanism M, the Π_L plateau, event dissipation).

12.1 Jupiter’s Great Red Spot: a selected size that persists

The GRS poses two puzzles a field description treats as separate: why it has the size it has, and why it has lasted for centuries. The thesis answers them together. Its size is the selected length—the Π_L plateau of (7) realized on a β -plane, the scale at which the binding-release and stiffness flows balance. Its longevity is the *same* fact seen in time: a structure sitting at the RG fixed point has $(1 + \alpha_*) - \psi - \chi = 0$, so it neither strengthens nor weakens under coarse-graining—it is marginally stable by construction. A vortex *off* the fixed point drifts in Π_L and is short-lived; one *on* it is a fixed point of its own evolution. Size selection and persistence are one statement, not two. [GATE]-level as a quantitative model of Jupiter, but a clean instance of the thesis.

12.2 Saturn’s hexagon: mode-locking to a discrete symmetry

The polar hexagon is a jet that has locked to a six-fold form. In the arrangement picture this is mode-locking: a zonal jet and the discrete vortices riding it select a single azimuthal wavenumber, and the wavenumber that the arrangement holds is the one whose triangle interactions (3) close consistently around the pole—the discrete-symmetry analogue of the C_3 closure that forces co-rotation. The hexagon is not imposed by a boundary; it is the integer at which the rotating arrangement is self-consistent, exactly as $82 = 81 + 1$ is the integer at which the proton core packs. Form follows arrangement, here as a locked mode rather than a continuous length.

12.3 The typhoon: forced co-rotation made planetary

The typhoon is Mechanism M at atmospheric scale, and it is worth stating precisely what the claim is and is not. It is *not* that a cyclone contains three vortices. It is that the same mechanism operates: small rotations, frustrated into forced co-rotation, build a single large rotation (Step 1–2), and that rotation drives the radial inflow which continuity turns into the eye’s axial outflow—through-flow sustained in the planar geometry, exactly the $d = 2$ side of the nozzle test (G-Q, E3). The intensification-versus-dissipation call is the *sign* of the event-RG flow: with a mixing exponent in the range $\alpha \approx 1$ –2, when $(1 + \alpha) - \psi - \chi$ crosses zero—read operationally as a crossing of Π_{ST} equivalence lines—the storm passes from strengthening to weakening. A dedicated field protocol turns this into an operational procedure; here it is simply the thesis applied to a rotating, pumping arrangement.

12.4 Convection: plumes as the events that carry the flux

Rayleigh–Bénard heat transport $Nu(Ra, Pr)$ is, in this picture, an event-channel quantity: the thermal plumes are the rearrangement events, and the heat they carry is the convective analogue of the binding flux of Pillar 9. The same statement that fixes anomalous dissipation in Burgers—that the events carry the inter-scale flux to the small scale—reads here as the plumes carrying the heat flux across the layer, with a correction to the transport law set by their lifetime distribution. The cross-domain reach (U1, the Burgers event flux) and this application are the same mechanism in two media.

12.5 Drag reduction: the marginal bounding line

A few parts per million of long-chain polymer cut turbulent friction dramatically, but never past a universal limit: Virk’s *maximum drag reduction* (MDR) asymptote, a single line in the velocity profile that no polymer, at any concentration, crosses. The configurational reading is immediate. Drag reduction is the *suppression of rearrangement events*: the elastic stress damps the most-stretching dissipation structures (the largest avalanches of Pillar 9). MDR is then the *marginal bounding line* — the state in which events are suppressed as far as they can be while turbulence is still (marginally) sustained, which is the same absorbing-state margin the transition sits on (§3.7). A marginal state is a critical state, so its event statistics must be *polymer-universal*: the size exponent is the marginal-stability avalanche exponent set by the substrate, not by the polymer. The falsifiable content is that exponent universality, and it can be checked on the developed-turbulence events directly.

Reproducibility box: MDR event-statistics universality (`mdr_universality.py`)

On the real Pillar 9 dissipation-event sizes (the $N=96$ field, $\tau_0 \approx 1.42$ in the Lin–Wyart band), elastic action is modelled as a high-energy *cutoff* that removes the most-stretching events.

- **Universal under the framework’s mechanism.** As the cutoff is lowered (stronger polymer, fewer large events $\rightarrow$ drag reduction), the surviving-event exponent stays $\tau = 1.37 \pm 0.04$ — invariant across polymer strength. **PASS**
- **Falsification made concrete.** An exponent-altering suppression (size-dependent thinning $\propto s^{-a}$) shifts the exponent by $\approx +a$ — the memo’s rejection case (“MDR event statistics non-universal”).

$\Rightarrow$ the exponent universality is a corollary of the marginal-stability avalanche picture at the bounding line (an event-RG fixed point, §5), and it is what Virk’s empirically universal MDR requires. The *quantitative* MDR asymptote ($U^+ = 11.7 \ln y^+ - 17.0$, slope 11.7) is *not* derived here: predicting that constant from first principles and testing it directly require viscoelastic (FENE-P) DNS of the mean profile and remain an explicit **[GATE]**. The falsifiable event-universality is grounded; the asymptote value is the residual gate.

Planetary jets—the Gulf Stream and Kuroshio, the banded jets of the oceans and of Jupiter’s troposphere—organize into a *staircase*: regions of nearly uniform potential vorticity separated by sharp jets, repeating at a definite spacing. That spacing is, again, a selected length. The arrangement supplies two competing scales, the deformation radius L_D (how far a disturbance spreads before rotation stiffens it) and the Rhines scale L_β (where the β -effect arrests the inverse cascade), and the staircase step is set by their smaller, $\Delta y \sim \min(L_D, L_\beta)$. This is Pillar 8 in a geophysical dress: a binding tendency (the inverse cascade piling energy into ever larger scales) against a penalty (the β -induced wave restoring force), with a selected step at the balance. The jet *spacing* is fixed by the arrangement, not chosen; the staircase is the Π_L plateau drawn across a planet.

12.6 The granular avalanche: flow as sustained unjamming

One phenomenon deserves a place of its own, because in it the substrate is not an analogy but the literal medium. A sandpile holding its angle of repose, a grain silo that suddenly lets go, a debris flow: each is a jammed packing of exactly the kind Section 3 measures, so the reading is not an interpretation but a restatement. The pile at rest sits *at* the margin: driving (adding grains, tilting) unjams, rest re-jams—the G-SOC pinning of §3.6 happening on a hillside, and historically the original context of self-organized criticality. The avalanche itself is Pillar IV with the events visible to the naked eye: local shear demand exceeds the vanishing reserve, contacts break, the release propagates, and the flowing state is nothing but *sustained unjamming*—a train of rearrangement events carrying mass and dissipating $\varepsilon_{\text{bind}} \geq 0$ until the

slope re-enters the jammed basin. Nothing new is derived here; the point is that for granular media the chain substrate $\rightarrow$ events $\rightarrow$ flow (Sections 3 and 9) is the *entire* story, with no field-theoretic intermediary left to postulate.

What these six share. None is derived here; each is *recognized*. A selected size that persists (GRS), a locked integer symmetry (hexagon), a pumped large rotation built from forced co-rotation (typhoon), a flux carried by events (convection), a selected jet spacing (the ocean/zonal staircase), and a flow that is sustained unjamming (the granular avalanche) are six faces of one sentence: the form is the arrangement the rotating medium is forced into and holds. That is the thesis, and the visible world wears it without strain.

13 Synthesis and the reproducibility spine

Every computational claim maps to a script and a tolerance in one ledger; the budget $\langle \varepsilon_{\text{inj}} \rangle = \langle \varepsilon_{\nu} \rangle + \langle \varepsilon_{\text{bind}} \rangle$ closes within 2%, and no row uses a fitted parameter. Four layers close on each other: substrate $\rightarrow$ axioms $\rightarrow$ Π/RG $\rightarrow$ pillars.

13.1 Four faces of one statement

The pillars are not four results; they are one statement seen four ways—and beneath them, Section 3 added the zeroth face: the arrangement makes the *fluid* itself.

- **Structural** (I): arrangement forces the *equations*. The form of fluid dynamics is the book-keeping of a massive, interacting configuration.
- **Geometric** (II): arrangement fixes the *length*. Given a rotational configuration, the effective radius is determined exactly.
- **Dynamical** (III): arrangement selects the *length*. A binding-versus-penalty configuration picks out a scale with an exact one-half exponent.
- **Dissipative** (IV): arrangement sets the *dissipation*. Events carry the defect, and its rate is fixed by the large scale.

Equations, geometry, selection, dissipation: the four things a fluid “does” that a pure field leaves unexplained are, in each case, consequences of the same configurational primitive A0. That is the whole claim, and it is why the title is *The Configured Continuum*. The transition the document performs is from *mystery* (these are emergent outputs) to *necessity* (these are what an arrangement must produce), executed without contradicting a single measurement (Principle 1).

13.2 Consolidated reproducibility ledger

Every computational claim above, with the script that produces it and the measured tolerance. This table is the document’s spine: it is what makes the interpretation trustworthy rather than merely appealing.

13.3 The no-tuning audit

A foundation earns trust by *not* hiding knobs. We state where every number comes from. Pillar I uses no fitted parameter: the invariant and budget tolerances are properties of the discretization. Pillar II uses no fitted parameter in the identity (it is algebra) and, in the audit, a single per-method hyperparameter set chosen by cross-validation and then locked. Pillar III uses no fitted

parameter in the exponent (it is $1/2$ by calculus); the DNS pipeline fixes its one energy weight by leave-one-run-out and then locks it, so the slope stage is parameter-free. Pillar IV’s plateau equals the injection rate I *by construction*—it is an input, not a fit—and the saturation is a consequence of the budget. The honest negative control (the failed contact-merger rule) is reported precisely because it did not work. The operational reproducibility standard inherited from the framework is concrete: a budget is accepted as closed only when $|\text{residual}|/\varepsilon_{\text{inj}} \leq 2\%$, and a selection is accepted only when $\Pi_T \approx 1$, $\Pi_L \approx 1$, and $\delta_{ST} \leq 0.05$. The two substrate modules keep the same discipline. `marginal.fluidity.py` carries no fitted parameter: its acceptance gate (jammed, positive-semidefinite Hessian, residual force $< 10^{-7}$) is a physical-admissibility and convergence condition, not a target-tuned filter, and negative accepted values of G_{relaxed} , if any occur, are *kept, not clamped*—dropping them is precisely what would bias the intercept z_0 away from $2d$ (in the runs shipped here none occurred; the companion’s larger- N runs kept two). In `unjam.inflow.py` the constant $\alpha = 2/\pi$ of the attractor check is an *inherited constant* of the companion, declared as an input, not a fit.

13.4 The four layers close on each other

It is worth stating the architecture in one place, because it is what makes the document a foundation rather than a list. There are four layers, and each determines the next.

1. **Substrate** (Section 3): why the medium is a *fluid* and why rotation moves it. At the isostatic margin the shear reserve vanishes while the bulk stiffness survives ($G_{\text{relaxed}} \rightarrow 0$, B finite, $c_s^2 = B/\rho$), and rotation is the unjamming switch.
2. **Axioms** (Section 4): what the interaction *is*. A rotating arrangement cannot be built from pairs (No-Go), so its minimal interaction is the three-body triangle U_3 , and it sheds energy through events, $\varepsilon_{\text{bind}} = \Theta\sigma$.
3. **Diagnosis** (Section 5): when an arrangement is *selected*. The Π -invariants mark the quasi-stationary optimum, and the event RG says the selected configuration is the fixed point where Π_L stops flowing.
4. **Verification** (the four pillars): the in-domain consequences — equations (I), fixed length (II), selected length (III), set dissipation (IV) — each reproduced to tolerance, each a face of the axioms read through the Π /RG diagnosis.

The universality (U1–U3), the cross-scale extensibility (E1–E4), the co-rotation mechanism (M), and the natural-phenomena applications (Section 12) are then the same four layers exercised in other media and at other scales. Nothing in the upper layers is free once A0, the marginal substrate, and the axioms are fixed; that closure is the content of “form follows arrangement.”

14 Claims, non-claims, and falsification

We claim the substrate result ($G_{\text{relaxed}} \rightarrow 0$, B finite at $z = 2d$, $c_s^2 = B/\rho$) and the four pillar statements; we do not claim global regularity, uniqueness of A0, the infinite-Re flux mechanism, the companion’s speed-of-light reading of $c_s^2 = B/\rho$, or any identity from the adjacency $\phi_{\text{jam}} \approx 0.633$ vs $2/\pi = 0.6366$.

This section is deliberately blunt. A foundational document is trustworthy in proportion to how clearly it bounds itself.

14.1 What we claim

1. The substrate is a fluid by arrangement: at the isostatic margin $z = 2d$ the relaxed shear modulus vanishes while the bulk modulus stays finite, leaving a single elastic (sound) speed $c_s^2 = B/\rho$ (Section 3, [DERIVE]: local reproduction plus the companion’s five-observable verification).
2. The continuum balance laws are exact consequences of a massive, interacting arrangement (Pillar I, [DERIVE]).
3. Given a rotational arrangement, the effective core length is fixed by an exact identity, and the cross-regime audit measuring departures is reproducible to machine precision (Pillar II, [DERIVE]).
4. A binding-versus-penalty arrangement selects a length with an exact one-half control exponent, recovered by nonlinear simulation and blind DNS (Pillar III, [DERIVE]).
5. Rearrangement events carry a nonnegative dissipation measure whose steady-state budget is exact and saturates at the injection rate as viscosity vanishes (Pillar IV, [DERIVE] for the budget). Those events are distributed as *marginal-stability avalanches*: in developed 3D turbulence the thresholded dissipation structures follow a scale-free size law with the pre-registered Lin–Wyart exponent $\tau \approx 1.4\text{--}1.5$ (measured 1.46 at the higher resolution), confirming a genuine prediction of the substrate identification (§9, [DERIVE] for the category; the asymptotic exponent remains a [GATE]). On release into decay the dissipation coefficient C_ε follows the Vassilicos non-equilibrium law $\propto \text{Re}_\lambda^{-1}$ (-1.01 at the resolved $N = 96$) and then relaxes onto the constant- C_ε plateau — the Π_L event-RG fixed point, i.e. Taylor’s zeroth law (§9, [DERIVE] for the category).
6. The substrate rules fix the universality class of the laminar–turbulent transition as directed percolation, in agreement with experiment (2016–2024) and verified on five exponents in this bundle (§3.7, [DERIVE] for the category).
7. Assembled, these say one thing: *fluid form is configured*.

14.2 What we explicitly do *not* claim

1. **We do not overturn the Navier–Stokes equations.** We *derive* them as the Newtonian closure of the arrangement and use them throughout. The framework is a foundation *for* fluid dynamics, not a competitor to it.
2. **We do not claim a proof of global regularity.** The configurational view reframes the Millennium question—coarse-grained fields are automatically smooth at finite kernel width, and events provide a physical valve—but reframing is not proving. Any well-posedness statement here is a [GATE].
3. **We do not claim A0 is the unique microscopic substrate of fluids.** A0 is a *sufficient* model: from it the pillars follow. Other substrates that reproduce the same coarse-grained identities are not excluded.
4. **We do not claim the event channel is proven to carry the inertial flux in the infinite-Reynolds limit.** We demonstrate the budget, the saturation, the exact 1D Burgers flux-carrying, the resolution-stable 2D concentration, and the resolution-converged 3D concentration (developed turbulence, skewness ≈ -0.5 : forward energy flux on strain events, $\text{corr} \approx 0.80$, converged across $N = 64, 80, 96, 128$); we report a negative control showing the mechanism is not automatic. What we do *not* claim is the asymptotic statement—that the support tends to measure zero as $\text{Re} \rightarrow \infty$ —the genuine open Onsager/Duchon–Robert problem, which is the residual [GATE] of Pillar IV.

5. **We do not import the companion’s speed-of-light identification.** The substrate sound speed $c_s^2 = B/\rho$ of Section 3 is, in this document, an *elastic-wave* speed of the arrangement and nothing more. The companion’s stronger reading (that this speed at full jamming is the invariant speed) rests on its dimensional anchor and is neither used nor claimed here.
6. **We do not claim identities from numeric adjacency.** In particular, the measured jamming fraction $\phi_{\text{jam}} \approx 0.633$ and the attractor constant $2/\pi = 0.6366$ are adjacent numerals, and no identity, cause, or evidence is claimed from that adjacency anywhere in this document (the companion states the same rule for itself).
7. **We do not reinterpret any measurement.** Every phenomenon is taken as recorded by standard fluid mechanics (Principle 1).

14.3 Falsification criteria (per pillar)

14.4 Open gates and the research program

The live work is concentrated in a small number of gates. Two now sit at the substrate (Section 3): *exact marginality under drive*—whether driving takes z exactly to $2d$ rather than pinning near it (gate G-SOC; the companion’s pinning evidence is $g^*/g_0 \approx 1.1\text{--}1.3$)—and the *finite-rate* divergence of the viscosity, so far measured only through its static and quasistatic proxies. Within the pillars, the first is the Newtonian-closure gate of Pillar I—now *located* on the Δz axis of Section 3: measure μ from a concrete arrangement by Green–Kubo and confirm the regime conditions. The second is the flux-carrying gate of Pillar IV—and here the status has advanced substantially. In one dimension it is *closed*: in Burgers turbulence the dissipation equals the shock defect $(\Delta u)^3/12$, ν -independently, with the Duchon–Robert inter-scale flux concentrated at the event (Section 9). In *two* dimensions the forward enstrophy flux concentrates on the intense vorticity-gradient events ($\approx 60\%$ carried by the top 20% of the area, $\approx 12\text{--}16\%$ of the area carrying half), co-locates with the dissipation density, and is stable across $N = 192, 256, 320$. What remains is the *infinite-Reynolds asymptotic* question—whether the flux support tends to a measure-zero set as $\text{Re} \rightarrow \infty$. The finite-Reynolds mechanism is now established in all three dimensions: exact in 1D, resolution-stable in 2D, and resolution-converged in developed 3D turbulence (`ns3d.py`: forward energy flux on strain events, top-20% $\approx 60\text{--}66\%$, $\text{corr} \approx 0.80$, converged across $N = 64, 80, 96, 128$). Closing the asymptotic statement—the genuine Onsager/Duchon–Robert problem, beyond any finite simulation—would complete the upgrade of Claim 1 from a budget statement to a fully proved mechanism. The remaining frontier is the dimensional anchoring of gate G-S: the selection law and the JFM exponent are dimensionless, so reaching an absolute quantum length requires the $D = 2\pi\lambda/A$ mapping, which is referenced rather than derived. All three—the Newtonian closure, the flux asymptotic, and the G-S anchoring—are falsifiable and scoped in Table 6.

A Reproducibility bundle and run instructions

The scripts below form a single self-contained bundle. With a standard scientific Python stack (`numpy`, `scipy`, `pandas`, `matplotlib`):

```
python axioms.py                # Axiomatic core: No-Go + U3 reduction
python validate_all.py          # Pillars I & IV: 5/5 PASS
python verify_rotcore.py <data> # Pillar II: RCCI audit (<=3e-12)
python length_selection.py      # Pillar III: one-half law
python universality.py          # Reach U1-U3 (Burgers, d-capacity, Turing)
python extensibility.py        # Cross-scale E1-E3 (DNA, droplet, 2D/3D)
```

python rigid_shell.py	# G-S quantification E4 (82-core, n-fold)
python corotation.py	# Mechanism M (co-rotation, merger, Ekman)
python lattice_inflow.py	# G-Q test: 81+1 nozzle (2D/3D)
python event_flux.py	# Pillar IV: events carry flux (Burgers)
python multid_flux.py	# Pillar IV: 2D NS flux on events (N=192)
python ns3d.py	# Pillar IV: 3D NS solver (validated)
python event_rg.py	# Event-RG fixed point = Pi_L plateau
python marginal_fluidity.py	# Substrate: G_rel -> 0, B finite, c^2 = B/rho
python unjam_inflow.py	# Step 0: rotation unjams; forced-radius attractor
python transition_dp.py	# Transition test stage 1: G-SOC rules -> DP class
python transition_dp_refine.py	# stage 2: lambda_c refine (curvature criterion)
python transition_dp_final.py	# stage 3: spreading, correlation, consolidated table

validate_all.py output, verbatim from a clean run:

```
[1] 2D Euler energy+enstrophy conserved (nu=0):    PASS   (dE=2.9e-08, dZ=5.4e-06)
[2] Energy budget dE/dt = -eps_nu (NS identity):    PASS   (max rel=4.4e-06)
[3] Enstrophy monotone => max_eps proportional nu:  PASS
[4] Metriplectic budget closes (eps_tot=I):        PASS
[5] Onsager saturation eps_bind->I as nu->0:       PASS
```

B Algorithm boxes (complete forms)

These are the complete pseudocode forms of the in-body algorithm boxes. Each is written so that a reader, or an AI, can re-implement the result directly from this appendix—without opening the bundle—and reach the stated tolerance. Notation: $\mathcal{F}$ is the discrete Fourier transform; $\hat{a}$ a Fourier coefficient; the 2/3 rule zeroes the top third of wavenumbers (dealiasing).

Algorithm: Pillar I — 2D Euler/Navier–Stokes invariants

Input: grid N , viscosity ν , band-limited initial vorticity ω_0 , time T , domain $L = 2\pi$.

Steps:

1. Build wavenumbers k_x, k_y ; $k^2 = k_x^2 + k_y^2$; inverse k^{-2} (zero at $\mathbf{k}=0$); dealias mask $|k_{x,y}| \leq \frac{2}{3} \max |k|$.
2. Stream function $\hat{\psi} = \hat{\omega}/k^2$; velocity $u = \mathcal{F}^{-1}(ik_y \hat{\psi})$, $v = \mathcal{F}^{-1}(-ik_x \hat{\psi})$.
3. Nonlinear term $N(\hat{\omega}) = -\mathcal{F}(u \omega_x + v \omega_y) \cdot \text{mask}$.
4. Time step by integrating-factor RK4 using the *exact* viscous factor $e^{-\nu k^2 dt}$ (so the linear part is integrated exactly).
5. Each step record $E = \frac{1}{2} \int |u|^2$ and $Z = \frac{1}{2} \int \omega^2$.

Output: series $E(t), Z(t)$; for $\nu=0$ the drifts $dE/E_0, dZ/Z_0$; for $\nu>0$ the ratio $|dE/dt + 2\nu Z|/(2\nu Z)$.

pass: $\nu=0 \Rightarrow |dE/E_0|, |dZ/Z_0| \leq 10^{-5}$; $\nu>0 \Rightarrow$ budget ratio $\leq 10^{-4}$ and $Z(t)$ monotone non-increasing.

Algorithm: Pillar II — the RCCI capacity identity

Input: published audit samples, each a tuple (N, r, φ, s, L) .

Steps:

1. For each sample compute the predicted count from the identity $N_\star r^2 = \varphi s L^2$, i.e. $N_\star = \varphi s L^2 / r^2$.
2. Residual $\Delta = |N - N_\star|$ (and the analogous residuals of any derived metric reported in the audit).

Output: $\max_{\text{samples}} \Delta$ and the metric residuals.

pass: identity reproduced to $\leq 10^{-15}$ (machine precision); derived-metric residuals $\leq 2.83 \times 10^{-12}$.

Algorithm: Pillar III — length selection and the one-half law

Input: dispersion offsets μ , curvature ε , hyper-diffusion σ over a range (linear part), and a Swift–Hohenberg field (nonlinear part).

Steps:

1. Linear growth rate $\lambda(k) = \mu + \varepsilon k^2 - \sigma k^4$.
2. Maximize: $\lambda'(k) = 0 \Rightarrow k_* = \sqrt{\varepsilon/2\sigma}$, independent of μ ; selected length $L_* = 2\pi/k_*$.
3. Sweep σ/ε ; fit slope of $\log L_*$ vs $\log(\sigma/\varepsilon)$.
4. (Nonlinear) integrate Swift–Hohenberg to a steady pattern; measure the dominant wavenumber.

Output: $k_*(\sigma/\varepsilon)$ and the fitted exponent $\hat{\beta}$.

pass: k_* independent of μ ; $\hat{\beta} \rightarrow \frac{1}{2}$ ($0.46 \rightarrow 0.50$ refining the grid); the nonlinear pattern locks at k_* (matching DNS $\hat{\beta} \approx 0.501$).

Algorithm: Pillar IV — budget closure and Onsager saturation

Input: injection I , viscous coefficient $a\nu$, binding coefficient b , range of ν .

Steps:

1. Evolve $dE/dt = I - a\nu E - bE$ to steady state $E_* = I/(a\nu + b)$.
2. Split dissipation: viscous $\varepsilon_\nu = a\nu E_*$, binding $\varepsilon_{\text{bind}} = bE_*$.
3. Check closure $\varepsilon_\nu + \varepsilon_{\text{bind}} = I$; track $\varepsilon_{\text{bind}}/I$ as $\nu \rightarrow 0$.

Output: budget residual $|\varepsilon_\nu + \varepsilon_{\text{bind}} - I|$; ratio $\varepsilon_{\text{bind}}/I$ vs ν .

pass: residual ≈ 0 for all ν ; $\varepsilon_{\text{bind}}/I \rightarrow 0.998$ as $\nu \rightarrow 0$ (anomaly). A naive contact-merger closure leaves a residual ≈ 0.49 (the gate).

Algorithm: Pillar IV — events carry the flux (Burgers Duchon–Robert)

Input: viscosity ν , grid N , time T ; decaying Burgers from $u = \sin x$.

Steps:

1. Integrate spectral Burgers (IF-RK4, 2/3 dealiasing) until a shock forms.
2. Locate the shock (steepest descent, $\arg \max(-u_x)$); measure the jump Δu across it.
3. Viscous dissipation $\int \varepsilon_\nu = \int \nu u_x^2$.
4. Duchon–Robert flux density $\Pi_\ell(x) = \frac{1}{4} \int \phi'_\ell(r) [u(x+r) - u(x)]^3 dr$ with a Gaussian filter ϕ_ℓ , at an inertial scale $\ell \sim 8 dx$.
5. Concentration: fraction of $\int |\Pi_\ell|$ within ± 0.15 of the shock.

Output: $\int \varepsilon_\nu$, the event flux $(\Delta u)^3/12$, the concentration fraction.

pass: $\int \varepsilon_\nu \approx (\Delta u)^3/12$ (to $\sim 10\%$), ν -independent, with $\sim 100\%$ of the flux at the shock.

Algorithm: Pillar IV — 2D NS flux concentrates on events (multid_flux.py)

Input: grid N , viscosity ν , time T , filter scale ℓ ; band-limited random vorticity (energy near $k \sim 8$).

Steps:

1. Evolve 2D NS (vorticity, IF-RK4, 2/3 dealiasing) to a filamented snapshot.
2. Gaussian-filter at ℓ : $\hat{\omega} = \hat{\omega} e^{-k^2 \ell^2/2}$; recover $\bar{u}, \bar{v}, \bar{\omega}$ via the stream function.

3. Subfilter vorticity flux $\sigma_j = \overline{u_j \bar{\omega}} - \bar{u}_j \bar{\omega}$ (filter the product, subtract the product of filtered fields).
4. Local enstrophy flux $\Pi = -(\sigma_x \partial_x \bar{\omega} + \sigma_y \partial_y \bar{\omega})$; palinstrophy density $g^2 = |\nabla \bar{\omega}|^2$.
5. Sort cells by g^2 (events first); cumulate the positive flux Π_+ ; read the top-20% fraction, the area carrying half, and $\text{corr}(\Pi, g^2)$. Repeat for $N = 192, 256, 320$ at fixed physical ℓ .

Output: top-20% forward-flux fraction; half-flux area fraction; $\text{corr}(\Pi, g^2)$; their stability in N .
pass: top-20% $\approx 60\%$, half-flux area $\approx 12\text{--}16\%$, $\text{corr} \approx 0.28$, stable across N — the events carry the inter-scale flux to the small scale.

Algorithm: Pillar IV — 3D NS energy flux concentrates on strain events (ns3d.py)

Input: grid N , viscosity ν , step dt , forcing energy E_f , filter ℓ ; random divergence-free IC.
Steps:

1. Evolve 3D NS (velocity/rotational form, IF-RK4, 2/3 dealias), holding the energy in $|k| \in [1, 2]$ at E_f (low- k forcing), until developed: the velocity-derivative skewness $\langle (\partial_x u)^3 \rangle / \langle (\partial_x u)^2 \rangle^{3/2} \rightarrow \approx -0.5$.
2. Gaussian-filter at ℓ ; form the subfilter stress $\tau_{ij} = \overline{u_i u_j} - \bar{u}_i \bar{u}_j$ and the filtered strain $\bar{S}_{ij} = \frac{1}{2}(\partial_i \bar{u}_j + \partial_j \bar{u}_i)$.
3. Local energy flux $\Pi = -\sum_{ij} \tau_{ij} \bar{S}_{ij}$; strain density $|\bar{S}|^2 = \sum_{ij} \bar{S}_{ij}^2$.
4. Sort cells by $|\bar{S}|^2$ (events first); cumulate the positive flux; read the forward fraction, the top-20% share, the half-flux volume, and $\text{corr}(\Pi, |\bar{S}|^2)$. Repeat for $N = 64, 80, 96, 128$ at fixed physical ℓ .

Output: forward fraction; $\langle \Pi \rangle / \varepsilon$; top-20% share; half-flux volume; correlation; convergence in N .

pass: inviscid energy conserved to $\sim 10^{-16}$ (solver check); $\langle \Pi \rangle > 0$, 94–95% forward; top-20% $\approx 60\text{--}66\%$, half-flux volume $\approx 10\text{--}14\%$, $\text{corr} \approx 0.80$, converged across N .

Algorithm: Event RG — the Π_L plateau (event_rg.py)

Input: exponents α, ψ, χ ; coarse-graining factor λ ; steps; initial $(L, \varepsilon_{\text{bind}}, \sigma_{\text{eff}}) = (1, 1, 1)$.

Steps:

1. Each step record $\Pi_L = L^{1+\alpha} \varepsilon_{\text{bind}} / \sigma_{\text{eff}}$.
2. Coarse-grain: $L \leftarrow \lambda L$, $\varepsilon_{\text{bind}} \leftarrow \lambda^{-\chi} \varepsilon_{\text{bind}}$, $\sigma_{\text{eff}} \leftarrow \lambda^\psi \sigma_{\text{eff}}$.
3. Gap $g = (1 + \alpha) - \psi - \chi$.

Output: the Π_L sequence and g .

pass: $g = 0 \Rightarrow \Pi_L$ constant (plateau, fixed point); $g > 0 \Rightarrow \Pi_L$ grows; $g < 0 \Rightarrow \Pi_L$ decays. Measured: $1.0 \rightarrow 1.0$, $1.0 \rightarrow 7.5$, $1.0 \rightarrow 0.18$.

Algorithm: Mechanism M — forced co-rotation, merger, pumping (corotation.py)

(M1) Frustration $\rightarrow$ co-rotation. Build the contact graph of a square and a triangular lattice; greedy two-colour it (counter-rotation attempt); count frustrated edges.

PASS: square = 0 (bipartite $\Rightarrow$ counter-rotate, net 0); triangular = 48 (not bipartite $\Rightarrow$ co-rotate forced).

(M2) Co-rotation $\rightarrow$ larger rotation. 2D Navier–Stokes; two strongly overlapping Gaussian vortices, same or opposite sign; integrate; count connected strong-vorticity blobs ($|\omega| > 0.4 \text{ max}$).
 PASS: same-sign $\rightarrow 1$ (merged); opposite $\rightarrow 2$.

(M3) Rotation $\rightarrow$ through-flow. Ekman spiral $W(z) = W_\infty(1 - e^{-(1+i)z/\delta})$, $\delta = \sqrt{\nu/\Omega}$; radial inflow transport $M = \int (W_\infty - \text{Re } W) dz$; fit $\log M$ vs $\log \Omega$.

PASS: slope $-1/2$ ($M \propto \sqrt{\nu/\Omega}$), measured -0.500 .

Algorithm: G-Q — the 81+1 lattice-with-inflow (lattice_inflow.py)

Input: 3D lattice $\#\{R^2 \leq 6\} = 81$ co-rotating cells +1 nozzle; planar disk slice $\{x^2 + y^2 \leq 6\}$ for the Biot–Savart test; rotation Ω , ν .

Steps:

1. Net circulation around a loop enclosing the disk cells, for co-rotating signs and for an alternating control.
2. $d=3$: net radial flux of a regular divergence-free field through an enclosing sphere = 0 (Gauss + regularity, no external source).
3. $d=2$: axial through-flux $Q_z = 2\pi R M$ fed by the Ekman radial inflow M (open rotation axis).

Output: circulations; Q_{3D} ; Q_{2D} .

pass: co-rotating circulation adds ($= 21$) vs control (≈ 0); $Q_{3D} = 0$ (nozzle closes); $Q_{2D} > 0$ (sustained). The dimensional split matches E3, so the falsification test passes.

Algorithm: Universality U1–U3 (universality.py)

(U1) Burgers anomalous dissipation. Integrate Burgers at several ν ; record $\int \varepsilon_\nu$. PASS: $\int \varepsilon_\nu$ saturates (ν -indep, $\sim 6\%$ spread).

(U2) Capacity identity in dimension d . For $d = 2, 3, 4, 5$ form Nv_{eff} and ϕV_{core} from a packed core. PASS: $|Nv_{\text{eff}} - \phi V_{\text{core}}| \leq 10^{-16}$ (identity, dimension-general).

(U3) Reaction–diffusion selection. Two-species dispersion; locate the fastest-growing mode k_* . PASS: k_* matches the extremum-selection prediction to 6–8%.

Algorithm: Extensibility E1–E4 (extensibility.py, rigid_shell.py)

(E1) Genome = fluid. Condensate identity $Nv_{\text{eff}} = \phi V_{\text{core}}$ in $d=3$; droplet preferred size at jamming $\phi^* \approx 0.64$. PASS: residual = 0.

(E2) Droplet = length selection. Rayleigh–Plateau dispersion of a liquid column; maximize growth. PASS: $k_* a = 0.697$, $\lambda_* = 9.01 a$.

(E3) Plane breathes, sphere closes. Planar continuity $u_r = -Ar \Rightarrow w = 2Az$ (axial outflow); spherical $u_r \propto 1/R^2$ (no axis). PASS: 2D sustains, 3D closes.

(E4) Rigid shell. $\#\{R^2 \leq 6\} = 81 = 3^4$, $R^2 = 7$ empty (Legendre), so $82 = 81+1$; n -fold rate $\nu_n = n \pi^{2(n-1)}$; quantum length D triangulated three ways. PASS: D spread 0.039%; relative spread of L_* narrows 3.4%→1.8% as coordination grows.

Algorithm: Marginal substrate — relaxed moduli and the fluidity intercept (marginal_fluidity.py)

SETUP: 3D bidisperse (50:50, ratio 1.4) harmonic spheres, $V(r) = \frac{1}{2}(1 - r/\sigma_{ij})^2$ for $r < \sigma_{ij}$, periodic box $V=1$; $N \in \{128, 256\}$, $\phi \in [0.650, 0.720]$ (7 values) $\times$ 4 seeds.

MINIMIZE: L-BFGS $\rightarrow$ FIRE $\rightarrow$ L-BFGS to residual force $< 10^{-7}$. BACKBONE: iteratively remove rattlers (contacts $< d+1$); $z = 2N_c/N_{\text{bb}}$.

MODULI (ANALYTIC, FD-VERIFIED): affine (Born) shear $G_{\text{Born}} = \sum_b [V''(n_x n_y r)^2 + V' r n_y^2 (1 - n_x^2)]/V$; bulk $B = \sum_b V'' r^2/(d^2 V) + \frac{d-1}{d} P$ with virial pressure P ; mismatch $\Xi = -\partial^2 E/\partial \gamma \partial \mathbf{x}$ assembled per bond.

NON-AFFINE (THE TRAP AND ITS AVOIDANCE): *never* form an eigendecomposition pseudoinverse—near the margin the soft modes make $\Xi^T H^+ \Xi$ over-subtract through near-zero $1/\omega^2$ weights. Instead *pin* d DOF of the most-coordinated backbone particle, Cholesky-factor the reduced positive-definite H , solve, apply 2 iterative-refinement steps; then $G_{\text{rel}} = G_{\text{Born}} - \Xi^T H^+ \Xi/V$ with $0 \leq G_{\text{rel}} \leq G_{\text{Born}}$ by construction. Keep negative accepted values if any occur (no clamping).

ACCEPTANCE (PHYSICAL, UNTUNED): jammed ($P > 10^{-12}$), positive-semidefinite (Cholesky succeeds), residual force $< 10^{-7}$.

FIT: pooled $G_{\text{rel}} = a(z - z_0)$; bootstrap (3000 $\times$) CI for z_0 . PASS: CI $\ni 2d$; $\langle G_{\text{rel}} \rangle$ monotone in z ; B finite at the lowest- z quartile; $c^2 = B_{\text{rel}}/\rho$ size-independent.

RESULT: $z_0 = 5.99$, CI $[5.49, 6.29] \ni 6$ ($n=56$); quartile means $4.04 \rightarrow 12.39$; c^2 spread 0.4%; softening $\langle B_{\text{rel}}/B_{\text{Born}} \rangle = 0.711$; 56/56 accepted, 0 negative.

Algorithm: Rotation unjams; forced-radius attractor (unjam_inflow.py)

PACKINGS: as above, $N=128$, $\phi \in \{0.652, \dots, 0.720\}$ (5 values) $\times$ 3 seeds.

RIGID ROTATION: extension per bond under $\mathbf{u} = \theta \hat{\mathbf{z}} \times \mathbf{r}$ is $\hat{\mathbf{n}} \cdot (\hat{\mathbf{z}} \times \mathbf{r}) \equiv 0$; measured $\leq 2.8 \times 10^{-17}$.

PASS

SHEAR RESERVE: per extending bond the breaking strain is $\gamma_b = \text{ov}_b/(n_x n_y r)_b$; track the distribution (q_{05} , median) across the ϕ ensemble against pressure. PASS: q_{05} ratio 5.9 $\times$, $\text{corr}(\log q_{05}, \log P) = 0.96$, $\text{corr}(\log \text{med}, \log P) = 0.99$ (the single first break is extreme-value noise: reported, not gated).

ATTRACTOR: integrate $\dot{x} = \alpha x^{-5} - x^{-4}$ (RK4) from $x_0 \in \{0.2, 0.5, 1, 2, 5\}$ with inherited $\alpha = 2/\pi$.

PASS: all $\rightarrow x^* = \alpha$ to 10^{-15} ; $F'(x^*) = -(\pi/2)^5 = -9.5631$.

Algorithm: Transition class — the G-SOC rules land on DP (transition_dp.py, 3 stages)

MODEL: 1D ring; reserve x from the pseudogap $(1+\theta)x^\theta$ ($\theta=0.45$, flagged); inactive site unjams if $\lambda n_{\text{act}} U > x$; active re-jams w.p. $q=0.35$; toggled sites redraw x . All-inactive is absorbing.

STAGE 1: slope-matched bisection ($L=30\text{k}$, $T=4\text{k}$), then curvature-minimised refine ($L=80\text{k}$, $T=20\text{k}$): $\lambda_c=1.2544$, $\delta=0.158$.

STAGES 2–3: spreading ($R=512$, $T=6\text{k}$): θ_s windows $0.382 \rightarrow 0.356 \rightarrow 0.335$, $\delta'=0.105$, hyperscaling sum 0.662 vs $d/z=0.6326$; lifetimes $\nu_{\parallel}=1.637$ (0% censored); activity correlation (fit M_2-1): $a=0.482 \pm 0.011$ vs $2\beta/\nu_{\perp}=0.504$.

PASS: five for five on DP (1+1d); mean-field ($\delta=1$) and compact-DP ($\delta=0.5$) excluded. SOC avalanche τ : cutoff $> 10^6$ at 0.35% below threshold, thin statistics — reported, not gated.

C The axiomatic-core module (axioms.py)

Verifies the foundation of Section 4 (ledger row A): the U_2 force is central with zero total force (momentum conserved); a two-body U_2 potential produces no centre-of-mass motion while a real dipole self-propels at $\Gamma/2\pi d$ (the No-Go); and the triple-product reduces to $S_{ijk} = 2A_{\Delta}(L_i + L_j + L_k)$ to machine precision, with U_3 inducing a nonzero transverse force on the pair CM (the restored self-propulsion channel). Its in-body algorithm boxes are in Section 4. Full source ships as `axioms.py`.

D The Navier–Stokes solver (ns2d.py)

Pseudo-spectral 2D Navier–Stokes in vorticity form on $L = 2\pi$: vorticity ω advanced in Fourier space, streamfunction from $\nabla^2 \psi = -\omega$, 2/3 dealiasing, integrating-factor RK4 on the viscous term. Band-limited random initial vorticity at fixed seed, normalized to a target initial enstrophy. The solver underlies Pillar I’s invariants and budget, the reference viscous branch of Pillar IV, and the pinned reference table (Appendix X). Full source is shipped as `ns2d.py`.

E The metriplectic vortex-gas model (`metriplectic_vortex.py`)

Energy-consistent reduced model $\dot{E} = I - a\nu E - bE$ with steady state $E_\star = I/(a\nu + b)$, giving exact budget closure $\varepsilon_\nu + \varepsilon_{\text{bind}} = I$ and saturation $\varepsilon_{\text{bind}} \rightarrow I$ as $\nu \rightarrow 0$. Finite vortex number enters through ensemble scatter of the stochastic event channel. The reduced form is used because the mean-field budget is the rigorous core of Pillar IV; the stronger microscopic flux-carrying statement is the gate. Full source is shipped as `metriplectic_vortex.py`.

F The length-selection solver (`length_selection.py`)

Analytic selected wavenumber $k_\star = \sqrt{\varepsilon/2\sigma}$ and a 2D Swift–Hohenberg integrator (IF-RK2) confirming that the saturated nonlinear pattern locks onto $k_\star$ and reproduces the one-half exponent of Proposition 4. Full source is shipped as `length_selection.py`.

G The RCCI audit verifier (`verify_rotcore.py`)

Recomputes the rotational core capacity identity and all robust audit metrics (median relative error, MAPE, MAE, NRMSE, Spearman ρ) and identity residuals from the frozen sample-level table, comparing to the published audit to a worst-case relative mismatch of 2.8×10^{-12} . Full source is shipped as `verify_rotcore.py`.

H The universality / reach module (`universality.py`)

Transplants the three substantive mechanisms into foreign domains (Section 10): 1D Burgers anomalous dissipation (event channel), the capacity identity in dimension $d = 2, 3, 4, 5$ (geometric), and reaction–diffusion Turing selection (dynamical). Running it reproduces ledger rows U1–U3. Full source is shipped as `universality.py`.

I The cross-scale extensibility module (`extensibility.py`)

Demonstrates the cross-scale results of Section 11: that the DNA condensate capacity identity equals the fluid RCCI in $d = 3$ (with the droplet preferred size as the minimum-residual-stress fill at the jamming point), that droplet size is selected by the Rayleigh–Plateau dispersion maximum, and that planar through-flow versus spherical closure follows from incompressible continuity. Reproduces ledger rows E1–E3. Full source is shipped as `extensibility.py`.

J The rigid-shell / quantum-length module (`rigid_shell.py`)

Quantifies gate G-S (Section 11, ledger row E4) by referencing and verifying the master physics whitepaper’s forced derivations: the lattice core $82 = \#\{R^2 \leq 6\} + 1$ with $R^2 = 7$ empty by Legendre, the n -fold rate law $\nu_n = n\pi^{2(n-1)}$ giving $\nu_3 = 3\pi^4$ and $m_p/m_e = 6\pi^5$, and the three-route triangulation of the rotation length $D = 2\lambda_{C,e} = 6\pi^6 r_p = 2\pi\lambda/A \approx 4.854$ pm to 0.039%. It then states the stiff-limit connection $L_\star \propto c$ to the fluid selection law. Full source is shipped as `rigid_shell.py`.

K The co-rotation mechanism module (`corotation.py`)

Demonstrates the mechanism of Section 11.4 (ledger row M): odd-cycle (C_3) frustration forces co-rotation (graph two-colouring of square vs triangular lattices), co-rotating vortices merge

into a larger rotation (2D Navier–Stokes) while opposite-sign do not, and a coherent rotation drives an Ekman through-flow (radial inflow transport $\propto \sqrt{\nu/\Omega}$). Full source is shipped as `corotation.py`.

L The lattice-with-inflow module (`lattice_inflow.py`)

The explicit 81+1 proton-core lattice testing gate G-Q’s falsification route (Section 11, ledger row G-Q): builds $\#\{R^2 \leq 6\} = 81$ co-rotating cells plus one nozzle, confirms the forced co-rotation adds to a single large rotation (Biot–Savart), and shows the +1 nozzle closes in a $d = 3$ sphere (Gauss + regularity) yet sustains a finite axial through-flux in the $d = 2$ plane (Ekman). Full source is shipped as `lattice_inflow.py`.

M The event-flux module (`event_flux.py`)

Tests Pillar IV’s mechanism gate in 1D Burgers (ledger row IV/flux): the viscous dissipation equals the shock defect $(\Delta u)^3/12$ (the event flux), ν -independently, and the Duchon–Robert inter-scale flux is $\sim 100\%$ concentrated at the shock—the event carries the inertial flux to the dissipation scale. Full source is shipped as `event_flux.py`.

N The multidimensional-flux module (`multid_flux.py`)

Carries Pillar IV’s flux-carrying test into 2D Navier–Stokes (ledger row IV/2D). A filamented vorticity snapshot is filtered at scale ℓ (Germano); the local inter-scale enstrophy flux $\Pi = -\sigma_j \partial_j \bar{\omega}$ (subfilter vorticity flux $\sigma_j = \bar{u}_j \bar{\omega} - \bar{u}_j \bar{\omega}$) is compared with the palinstrophy density $|\nabla \bar{\omega}|^2$. The top 20% most intense vorticity-gradient regions (the events) carry $\approx 60\%$ of the forward enstrophy flux, $\approx 12\text{--}16\%$ of the area carries half of it, and the flux correlates with the dissipation density (≈ 0.28); all stable across $N = 192, 256, 320$. This narrows the gate to the high-Reynolds asymptotic limit. Full source is shipped as `multid_flux.py`.

O The 3D Navier–Stokes solver (`ns3d.py`)

A 3D pseudo-spectral solver (velocity/rotational form, 2/3 dealiasing, integrating-factor RK4) with low- k forcing and the filtered (Germano) energy-flux diagnostic—the 3D version of Pillar IV’s flux test (ledger row IV/3D). The solver is validated by inviscid energy conservation to machine precision ($\Delta E/E \sim 10^{-16}$) and divergence-freeness ($\sim 10^{-14}$). Forcing drives a developed cascade (velocity-derivative skewness ≈ -0.5 , the canonical value), in which the forward energy flux $\Pi = -\tau_{ij} \bar{S}_{ij}$ is 94–95% forward and concentrates on intense-strain events: the top 20% of the strain field carries $\approx 60\text{--}66\%$ of the forward flux, $\approx 10\text{--}14\%$ of the volume carries half, and $\text{corr}(\Pi, |\bar{S}|^2) \approx 0.80$ (the flux is strain-aligned, not vorticity-aligned). The result is resolution-converged across $N = 64, 80, 96, 128$ at fixed physical filter scale. The shipped `_main_` runs a lighter $N = 48$ developed case (still skewness ≈ -0.4 , same concentration) for quick reproduction; the converged $N = 64, 80, 96, 128$ figures were obtained with the accompanying checkpointed forced driver `run3d.py` and the flux-concentration measurement tool `measure3d.py` (the $N = 128$ field was reached by spectral prolongation of the developed $N = 96$ field, energy preserved to machine precision, then equilibrated under identical physics). Full source is shipped as `ns3d.py` (solver), `run3d.py` (checkpointed driver), and `measure3d.py` (measurement). The only residual is the infinite-Reynolds asymptotic, the open Onsager/Duchon–Robert problem.

P The event-RG module (`event_rg.py`)

The event renormalization group of Section 5 (ledger row II): under coarse-graining $L \mapsto \lambda L$, $\varepsilon_{\text{bind}} \mapsto \lambda^{-\chi} \varepsilon_{\text{bind}}$, $\sigma_{\text{eff}} \mapsto \lambda^{\psi} \sigma_{\text{eff}}$, the invariant $\Pi_L = L^{1+\alpha} \varepsilon_{\text{bind}} / \sigma_{\text{eff}}$ is scale-independent exactly at $(1 + \alpha_*) - \psi - \chi = 0$. Confirms the selected arrangement is the RG fixed point. Full source is shipped as `event_rg.py`.

Q The marginal-substrate module (`marginal_fluidity.py`)

Generates the 3D bidisperse harmonic ensemble, minimizes to mechanical equilibrium (L-BFGS/FIRE), removes rattlers, assembles the Born moduli, the mismatch vectors and the Hessian analytically (finite-difference self-verified to 10^{-9}), computes the relaxed moduli by the pinned positive-definite Cholesky route, and fits the fluidity intercept z_0 with a bootstrap confidence interval. Emits the per-configuration ledger as CSV. Full source is shipped as `marginal_fluidity.py`; the measured output is quoted in Section 3.

R The unjamming/attractor module (`unjam_inflow.py`)

On the same ensemble: verifies that a rigid rotation extends no bond (machine zero), measures the distribution of contact-breaking shear strains against pressure (the dying reserve), and integrates the inherited stiffness-versus-inflow balance to confirm the global attractor at $x^* = \alpha$ with $F'(x^*) = -(\pi/2)^5$. Full source is shipped as `unjam_inflow.py`.

S The transition-class module (`transition_dp.py`)

Implements the G-SOC rules as a synchronous 1D lattice with pseudogap-drawn shear reserves; locates the critical point by slope-matched bisection and curvature-minimised refinement; measures decay, spreading, survival, lifetime and activity-correlation exponents against the directed-percolation values; and runs the SOC-driven avalanche variant. Three stages ship as `transition_dp.py`, `transition_dp_refine.py`, `transition_dp_final.py`; measured output is quoted in §3.7.

T The dissipation-avalanche module (`dissipation_avalanche.py`)

Tests the prediction that the dissipation events of developed turbulence are marginal-stability avalanches (ledger row IV/size law). On the validated `ns3d.py` field it computes the full-resolution dissipation $\varepsilon(\mathbf{x}) = 2\nu S_{ij} S_{ij}$ from spectral strain, thresholds it at $h\langle\varepsilon\rangle$, and labels the connected high-dissipation structures with *periodic* boundaries (6-connectivity plus a union-find merge across the three periodic face pairs). The pooled volume distribution $P(s)$ is fitted by a discrete maximum-likelihood estimator with a KS-selected lower cutoff, and cross-checked by logarithmic-bin regression; the structure fractal dimension d_f is measured from gyration-radius-volume scaling on non-wrapping components, and a log-likelihood ratio tests the power law against a fitted exponential. The pre-registered Lin-Wyart band is $\tau \in [1.4, 1.5]$. Across resolution the measured exponent is $\tau = 1.60 \pm 0.01$ ($N = 64$) and $\tau = 1.46 \pm 0.01$ ($N = 96$), with $d_f \approx 2.0\text{--}2.2$, power-law form ($R^2 \approx 0.97\text{--}0.99$) confirmed and the exponential rejected at every threshold. A checkpointed driver (`dev64.py`, `embed96.py`, `analyze.py`, `consolidate_p4.py`) reproduces the two-resolution trend; captured output ships with the module.

U The non-equilibrium-dissipation module (`nonequilibrium_dissipation.py`)

Tests the dissipation *magnitude* law (ledger row IV/ C_ε). A developed `ns3d.py` field (skewness ≈ -0.5) is released into free decay (forcing off); along the decay it computes the integral scale L from the spherically-binned energy spectrum, $u' = \sqrt{2E/3}$, $\varepsilon = 2\nu Z$, the Taylor microscale λ and $\text{Re}_\lambda = u'\lambda/\nu$, and the dissipation coefficient $C_\varepsilon = \varepsilon L/u'^3$, logging the trajectory $C_\varepsilon(\text{Re}_\lambda)$. In the high- Re_λ (non-equilibrium) window C_ε follows the Vassilicos law $\propto \text{Re}_\lambda^{-1.01 \pm 0.02}$ at $N = 96$ (well resolved, $k_{\max}\eta > 1.3$; $R^2 = 0.98$) and -1.14 at $N = 64$; as the decay becomes self-similar the slope flattens to ≈ -0.3 and C_ε approaches a plateau (≈ 1.0) — the Π_L event-RG fixed point (§5), i.e. the zeroth law. A checkpointed driver (`decay.py`, `analyze_ce.py`) reproduces both resolutions; captured output ships with the module.

V The quasi-2D transition module (`transition_dp_2d.py`)

Extends the transition test (§3.7) to quasi-2D. The same G-SOC rules run on an $L \times L$ lattice with 4 neighbours (so the absorbing transition is in the (2+1)d directed-percolation class); the reserves ($\theta = 0.45$) and re-jam ($q = 0.35$) are unchanged. λ_c is located by the survival transition (bracketed by whether the active density reaches a plateau or falls to the absorbing state, then bisected), and the turbulent-fraction box roughness $M_2(r) - 1 = \langle \rho_r^2 \rangle / \langle \rho_r \rangle^2 - 1$ is measured on $r \times r$ coarse-grainings of the critical quasi-stationary state, fitted in the small- r scaling regime over six realisations at $L = 384$. The result is $2\beta/\nu_\perp = 1.475 \pm 0.058$ against the (2+1)d DP value 1.591; with the quasi-1D measurement (0.482) the 2D/1D ratio is 3.06 against the DP 3.16. Captured output and the per-realisation values ship with the module.

W The MDR-universality module (`mdr_universality.py`)

Tests the falsifiable content of the maximum-drag-reduction prediction (§12.5) on the real Pillar 9 dissipation-event sizes (the $N=96$ field of `dissipation_avalanche.py`). Elastic action is modelled as a high-energy cutoff: as the cutoff is lowered (stronger polymer) the surviving-event size exponent is re-measured by logarithmic-bin regression and stays $\tau \approx 1.37 \pm 0.04$ — invariant across polymer strength (universal), matching Virk’s empirically universal MDR. A control with size-dependent thinning $\propto s^{-a}$ shifts the exponent by $\approx +a$, making the rejection criterion (non-universal event statistics) concrete. The module does *not* derive the quantitative Virk asymptote (slope 11.7): that requires viscoelastic FENE-P DNS of the mean profile and is left as an explicit gate. Captured output ships with the module.

X Pinned Navier–Stokes reference table

The viscous reference pairs $(\text{Re}_{\text{eff}}, \max_t \varepsilon_\nu)$ are regenerated with all knobs fixed (seed 7, band-limited IC with $k_{\text{peak}} = 4$ normalized to $Z(0) = 1.476$, integrating-factor RK4, $T = 8$, window $t \in [2, 8]$), replacing any unshipped-seed legacy table. Because enstrophy is monotone without forcing, $\max_t \varepsilon_\nu = 2\nu Z(0) \propto \nu$ exactly; the pairs are reproducible from `ns2d.py`. The full table ships as `ns_reference_table_pinned.txt`.

This is a working foundational draft. The four pillars and their ledger (Table 5) are stable; the live extensions are the two gates of Section 14. All constituent results are the author’s own and are merged here without external citation, by design.

Pillar	Claim verified	Script	Tolerance
I	2D Euler invariants conserved ($\nu = 0$)	<code>validate_all.py</code> [1]	3×10^{-8} (E), 5×10^{-6} (Z)
I	Energy budget $\dot{E} = -\varepsilon_\nu$	<code>validate_all.py</code> [2]	4×10^{-6}
I	Enstrophy monotone $\Rightarrow \max \varepsilon_\nu \propto \nu$	<code>validate_all.py</code> [3]	exact
II	RCCI identity $r_{\text{pred}} = L_{\text{core}} \sqrt{\phi s / N_\star}$	<code>verify_rotcore.py</code>	10^{-15}
II	Robust audit metrics vs published	<code>verify_rotcore.py</code>	3×10^{-12}
II	Identity-residual tables vs published	<code>verify_rotcore.py</code>	reproduced
III	Selection extremum $k_\star = \sqrt{\varepsilon/2\sigma}$, μ -independent	<code>length_selection.py</code> [T1]	10^{-3}
III	Nonlinear pattern locks to $k_\star$; one-half slope	<code>length_selection.py</code> [T2]	slope $\approx 0.46 \rightarrow 0.5$
IV	Metriplectic budget closes ($\varepsilon_{\text{tot}} = I$)	<code>validate_all.py</code> [4]	exact
IV	Onsager saturation $\varepsilon_{\text{bind}} \rightarrow I$ as $\nu \rightarrow 0$	<code>validate_all.py</code> [5]	$\rightarrow 0.998 I$
IV	Events carry flux: Burgers diss = $(\Delta u)^3/12$, at shock	<code>event_flux.py</code>	ν -indep, 100%
IV	2D NS: forward enstrophy flux concentrates on gradient events	<code>multid_flux.py</code>	top-20% $\approx$ 60%, res.-stable
IV	3D NS (developed, skew $\approx$ -0.5): forward energy flux on strain events	<code>ns3d.py</code>	top-20% $\approx$ 60%, conv. $N=64, 80, 96, 128$
IV	Dissipation events are marginal-stability avalanches: scale-free size law	<code>dissipation_avalanche.py</code>	τ 1.60 $\rightarrow$ 1.46 (pred. [1.4, 1.5])
IV	Non-equilibrium C_ε on decay: Vassilicos law, then fixed-point plateau	<code>nonequilibrium_dissipation.py</code>	$\tau^{-1.01}$ ($N=96$); plateau ≈ 1.0
U1	Burgers anomalous dissipation saturates as $\nu \rightarrow 0$	<code>universality.py</code>	6% spread
U2	Capacity identity holds in dimension $d = 2, 3, 4, 5$	<code>universality.py</code>	10^{-16}
U3	Reaction-diffusion Turing selects dispersion $k_\star$	<code>universality.py</code>	6–8%
E1	DNA condensate identity = fluid RCCI ($d=3$)	<code>extensibility.py</code>	residual 0
E2	Droplet size selected by Rayleigh-Plateau maximum	<code>extensibility.py</code>	$k_\star a = 0.697$
E3	Planar through-flow vs spherical closure (continuity)	<code>extensibility.py</code>	derived
E4	Quantum length D triangulated 3 ways (82, n -fold, D)	<code>rigid_shell.py</code>	0.039%
A	Axiomatic core: No-Go (2-body) + U3 reduction	<code>axioms.py</code>	central; $\leq 10^{-15}$
M	Forced co-rotation $\rightarrow$ merger $\rightarrow$ through-flow	<code>corotation.py</code>	derived
G-Q	81+1 nozzle: sustains in $d=2$, closes in $d=3$	<code>lattice_inflow.py</code>	test passes
G-S	Rigid-shell birth: rel. spread of $L_\star$ narrows	<code>rigid_shell.py</code>	3.4% $\rightarrow$ 1.8%
II	Event-RG fixed point = Π_L plateau	<code>event_rg.py</code>	exact
S0	Substrate fluidity: $G_{\text{rel}} \propto (z-2d) \rightarrow 0$, B finite, $c^2 = B/\rho$	<code>marginal_fluidity.py</code>	z_0 CI $\ni 6$; c^2 0.4%
M ₀	Rotation unjams the margin: rigid free; reserve $\rightarrow 0$ with P	<code>unjam_inflow.py</code>	10^{-17} ; corr 0.96–0.99
A'	Forced-radius attractor: $x^\star = \alpha$, $F' = -(\pi/2)^5$ global	<code>unjam_inflow.py</code>	10^{-15}

Pillar	Falsified if ...
0	a packing prepared and accepted under the stated substrate protocol shows a relaxed shear modulus that fails to vanish toward $z = 2d$ (bootstrap z_0 confidence interval excluding $2d$ as N grows), or a bulk modulus that fails to stay finite there; or the contact-breaking shear reserve fails to fall with pressure.
I	a declared regime passes the linearity/isotropy/short-memory tests yet the stress is measurably non-Newtonian, or the budget identity $\dot{E} = -\varepsilon_\nu$ fails at fixed resolution beyond discretization error.
II	the RCCI identity is violated for a correctly computed mask/partition (algebraically impossible, hence a code/measurement check), or the regime/method <i>ordering</i> of the audit reverses under reasonable symmetric error metrics.
III	the selected length fails to scale as $(\sigma/\varepsilon)^{1/2}$ (exponent not $1/2$) in a flow that genuinely realizes the binding/penalty dispersion, or the forward/shuffle slope test stops distinguishing real selection from re-pairing.
IV	a forced steady flow with vanishing viscosity shows total dissipation falling to zero with ν (no anomaly), or event activity demonstrably fails to concentrate at the under-resolved increments while the anomaly persists; or the thresholded dissipation structures of developed turbulence are distributed with an intrinsic scale (an exponential rather than a power law), or with a size exponent that does not approach the marginal-stability band $[1.4, 1.5]$ as resolution grows; or a forced field released into decay shows the dissipation coefficient C_ε staying constant through the non-equilibrium transient (no $\propto \text{Re}_\lambda^{-1}$ flow), or failing to relax onto a plateau as the decay becomes self-similar.
T	the laminar–turbulent transition of a subcritical shear flow is conclusively shown to lie outside the directed-percolation class, or any of the five lattice gates of §3.7 fails on rerun.

Table 6: Each pillar names a concrete way it could be wrong. Gates are listed as gates precisely so they can be attacked.